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(54) **VISION SYSTEM FOR MARITIME VEHICLE**

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H04N 23/54 (2023.01)

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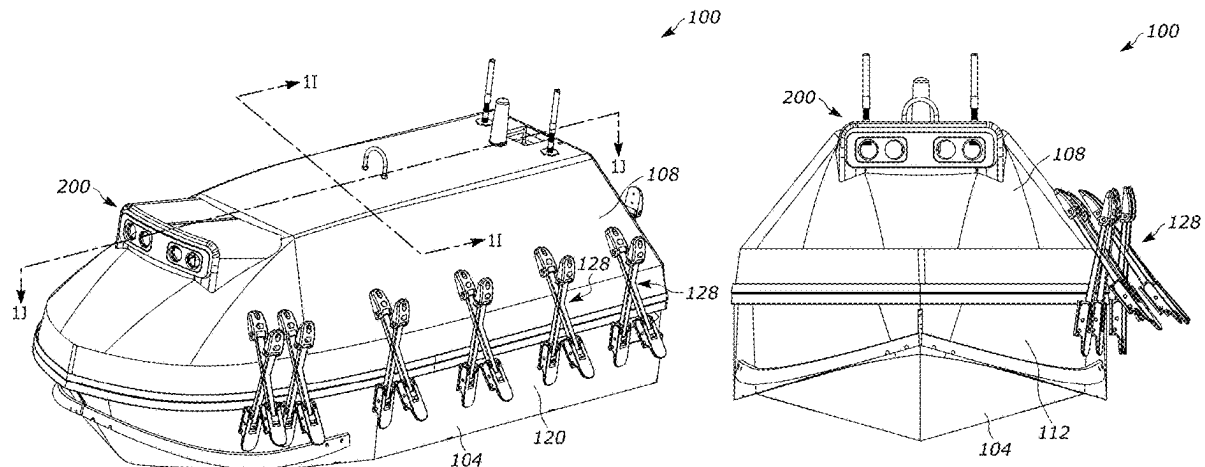
H04N 13/204 (2018.01)

H04N 23/20 (2023.01)

(57)

ABSTRACT

A camera module for a maritime vehicle. The camera module includes a frame comprising first and second openings, first and second windows disposed in the frame immediately adjacent the first and second openings, respectively, a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings, a stereo camera disposed within the frame, the stereo camera disposed immediately adjacent the first opening, and an infrared (IR) camera disposed within the frame, the IR camera disposed immediately adjacent the second opening.



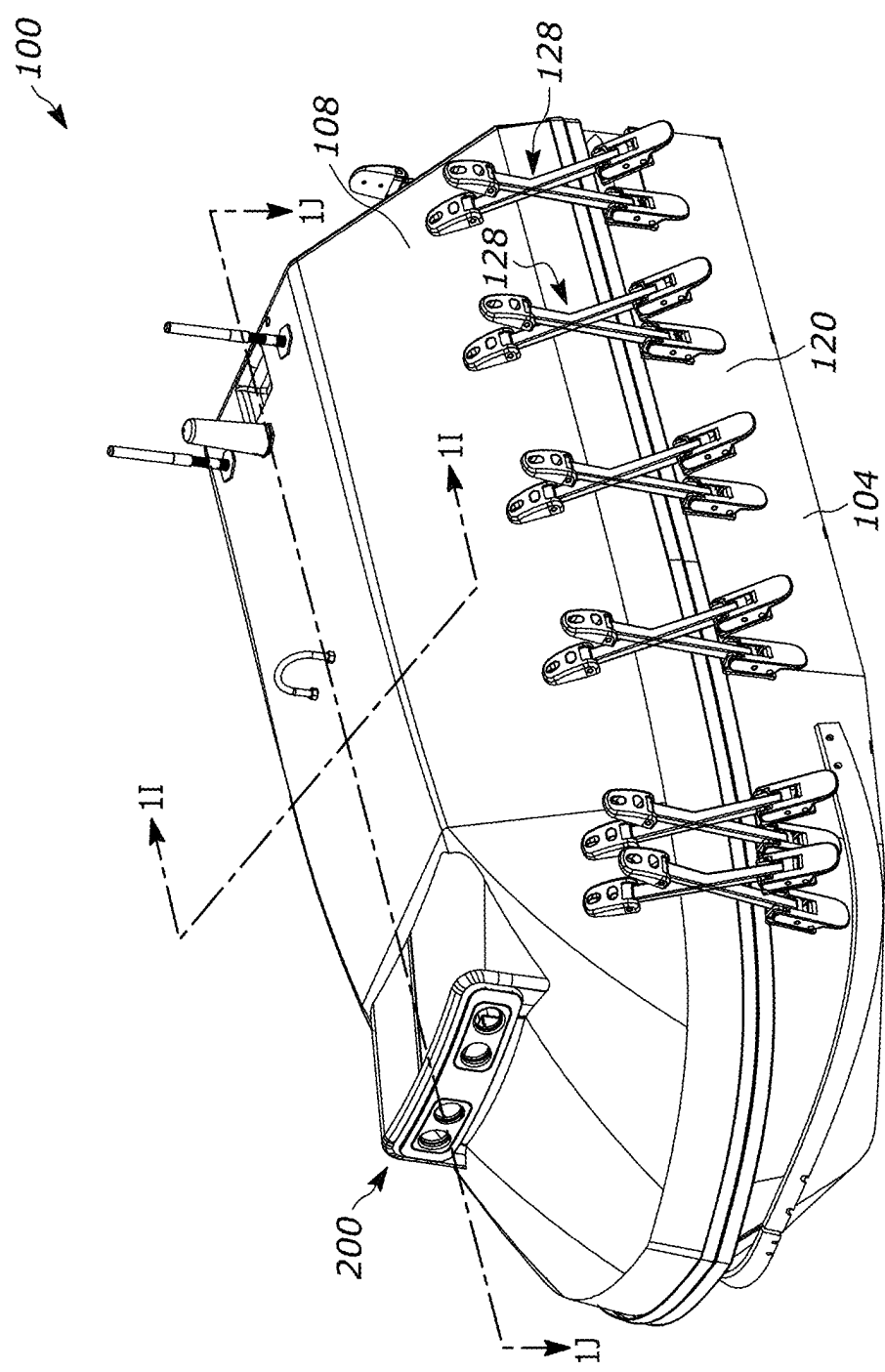


FIG. 1A

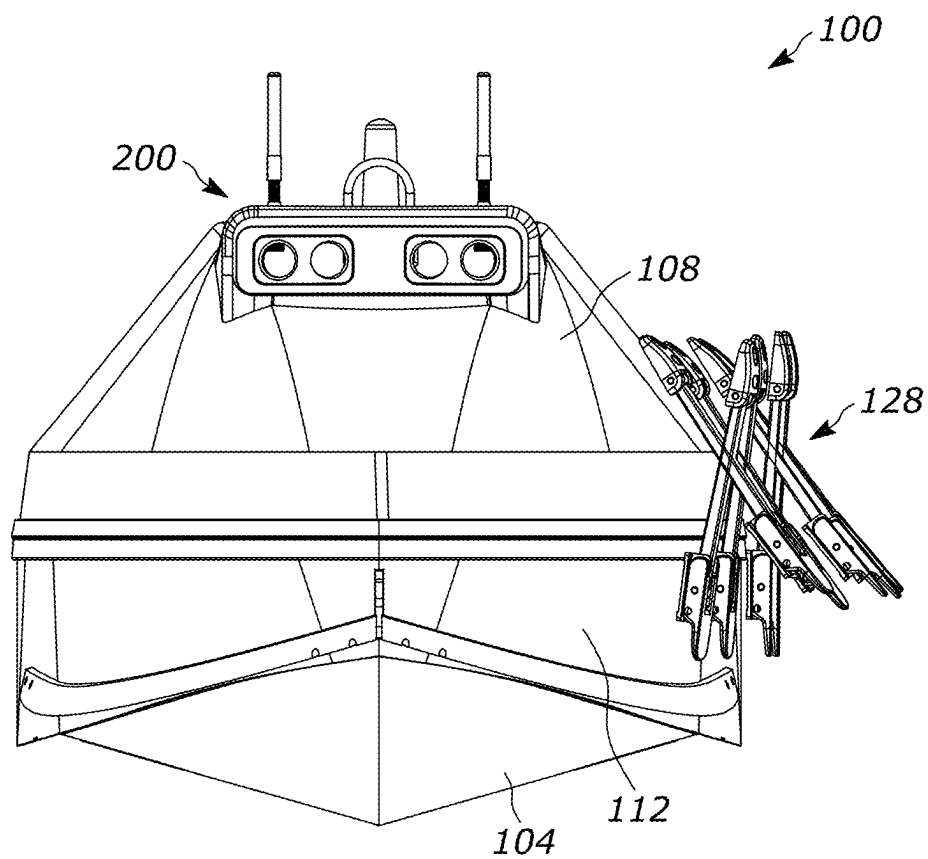


FIG. 1B

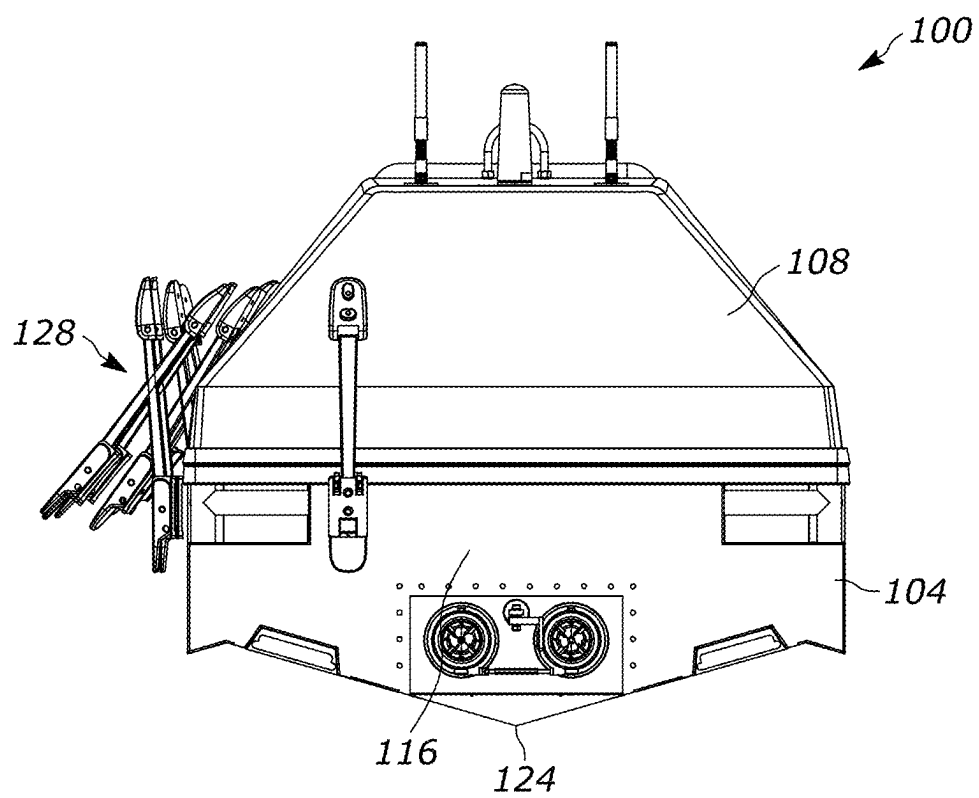


FIG. 1C

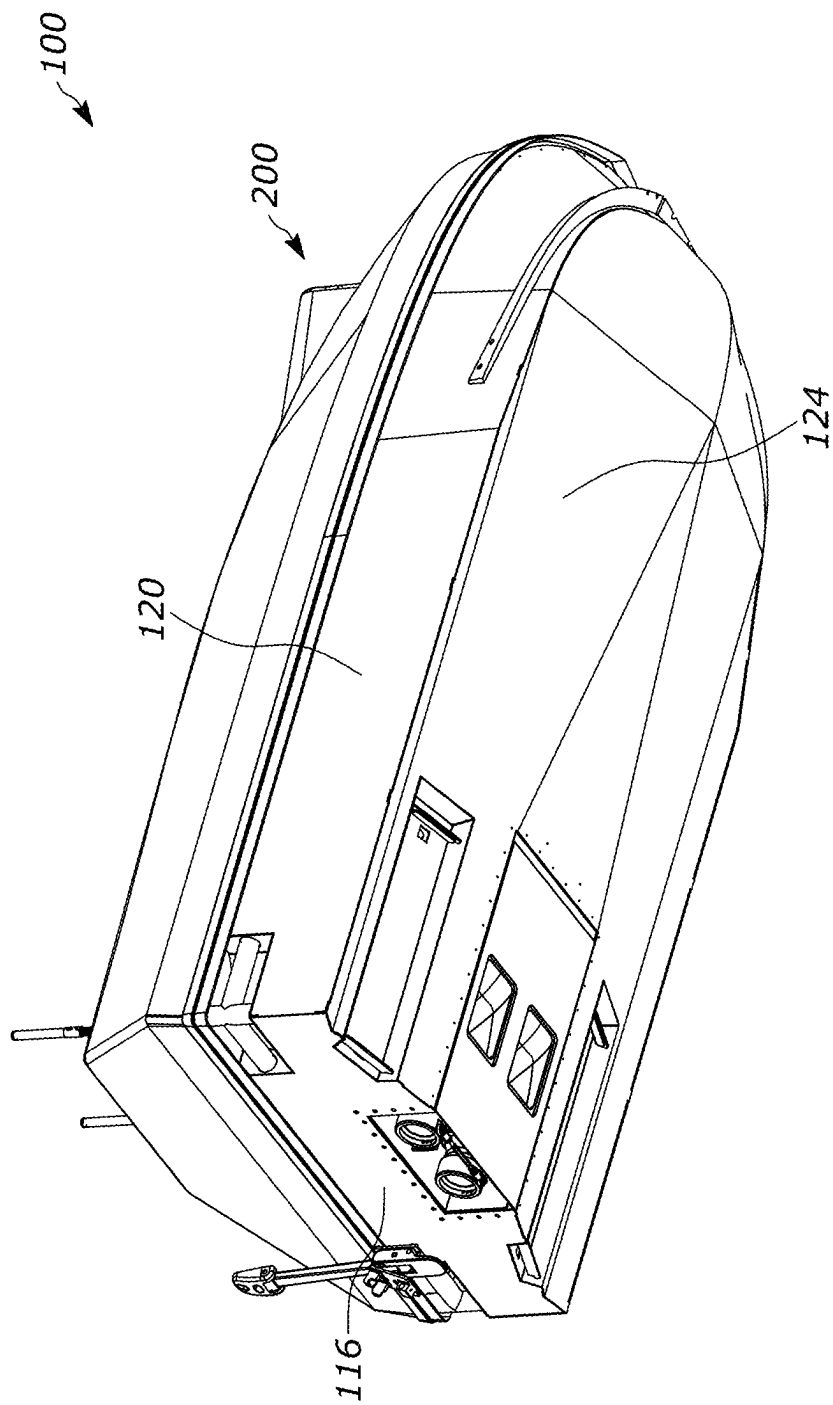


FIG. 1D

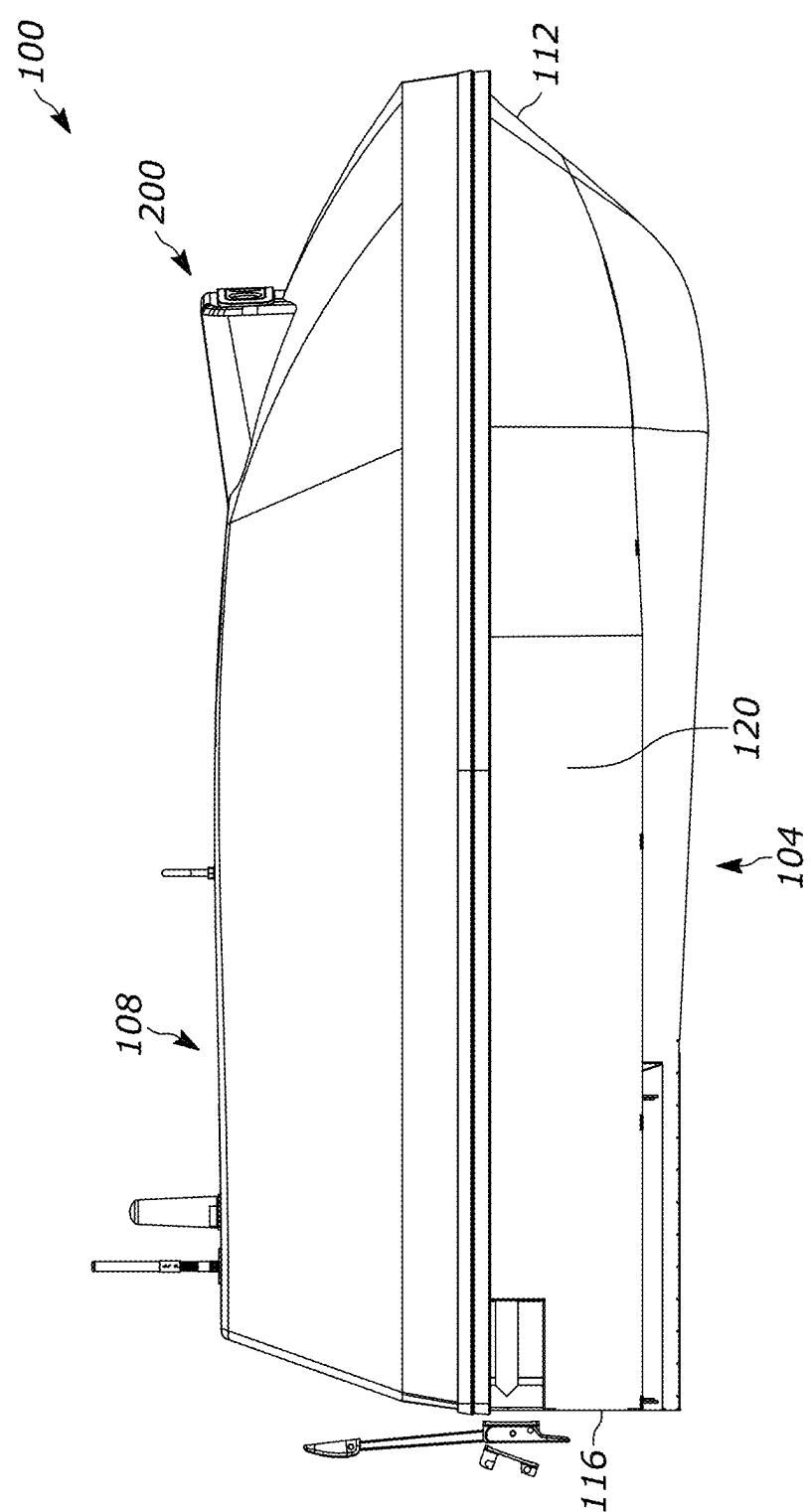


FIG. 1E

FIG. 1F

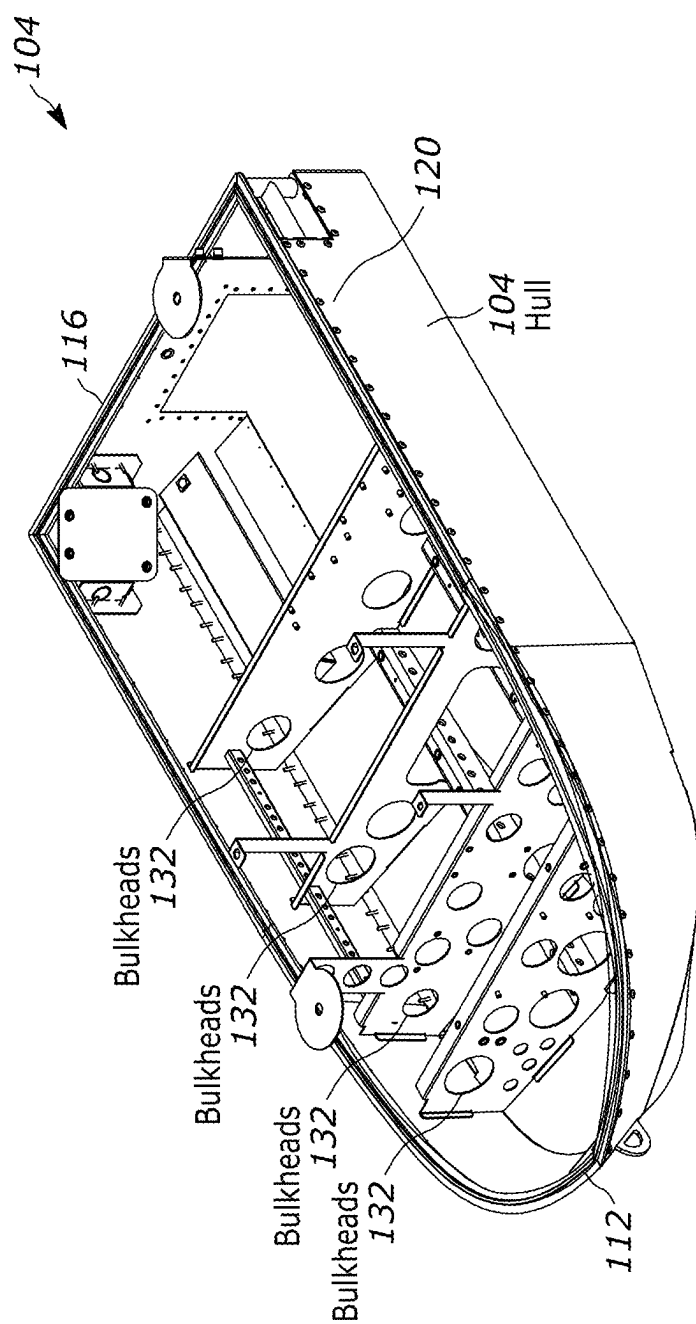


FIG. 1G

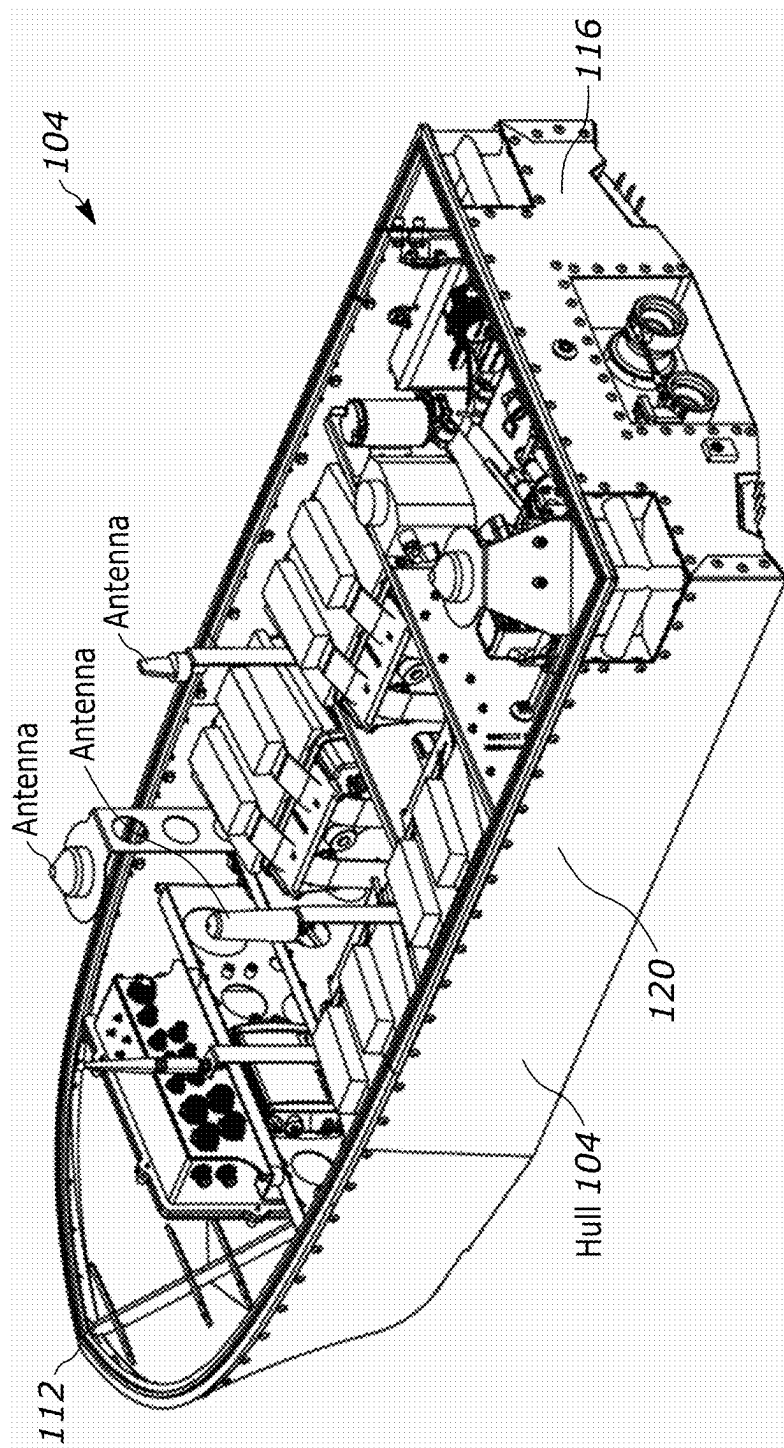


FIG. 1H

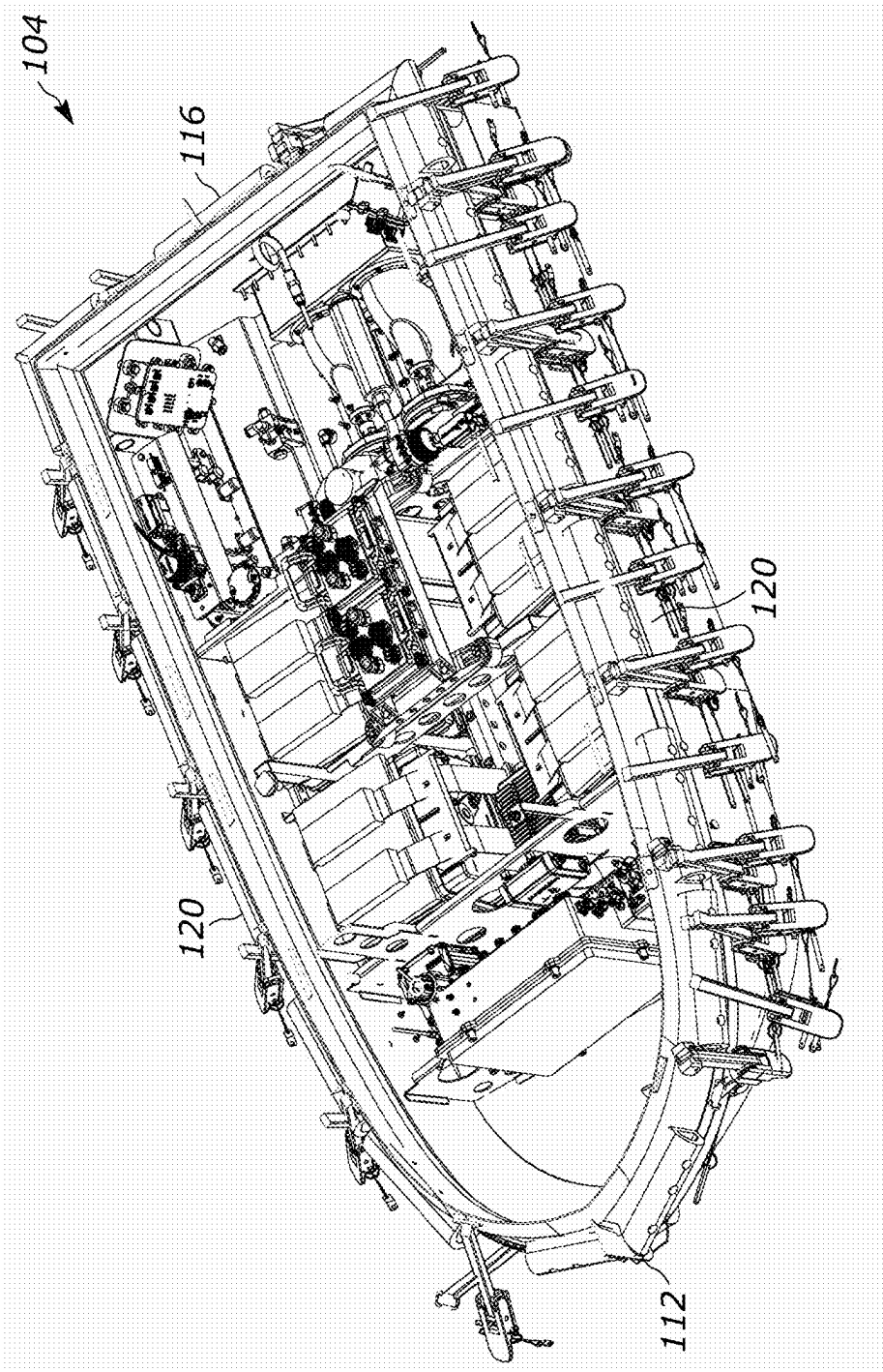


FIG. 11

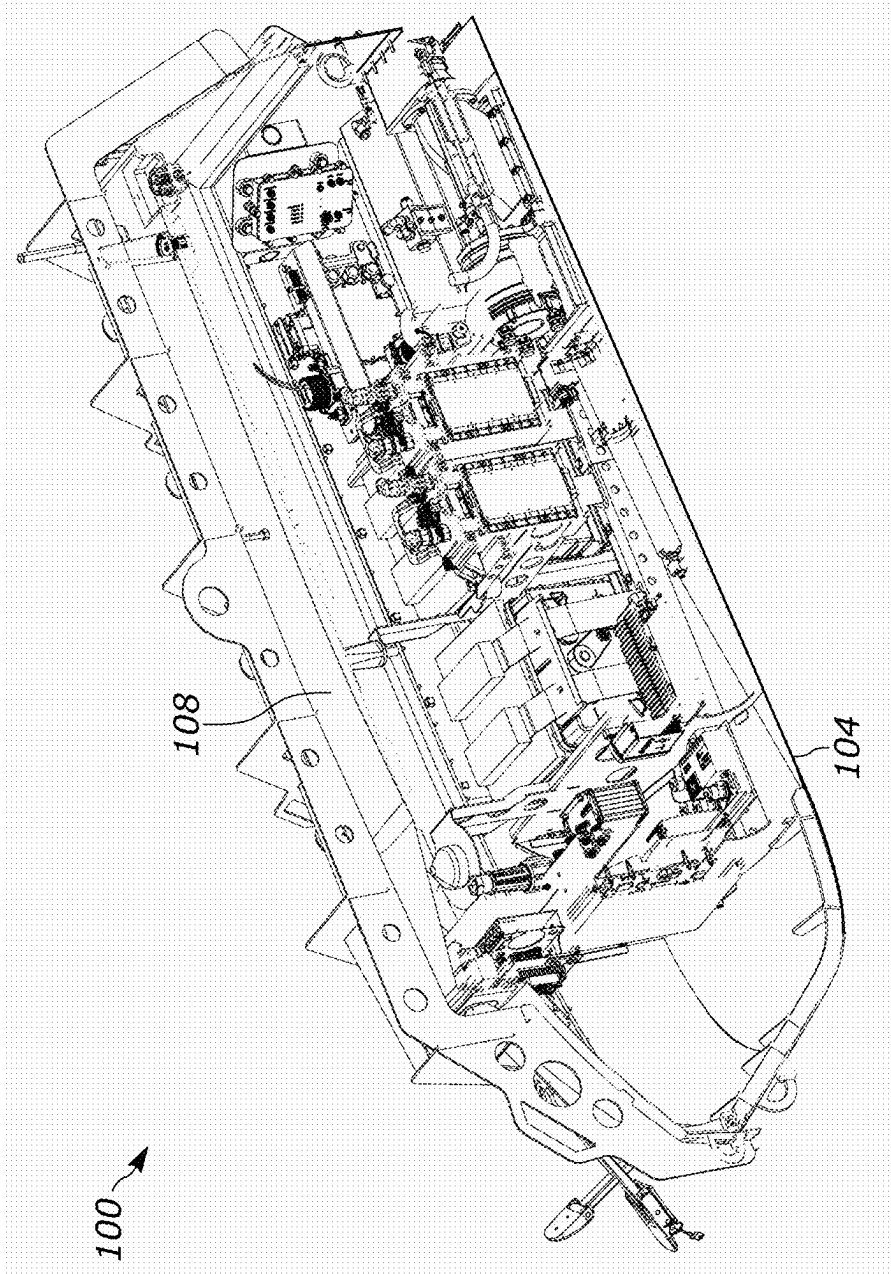


FIG. 1J

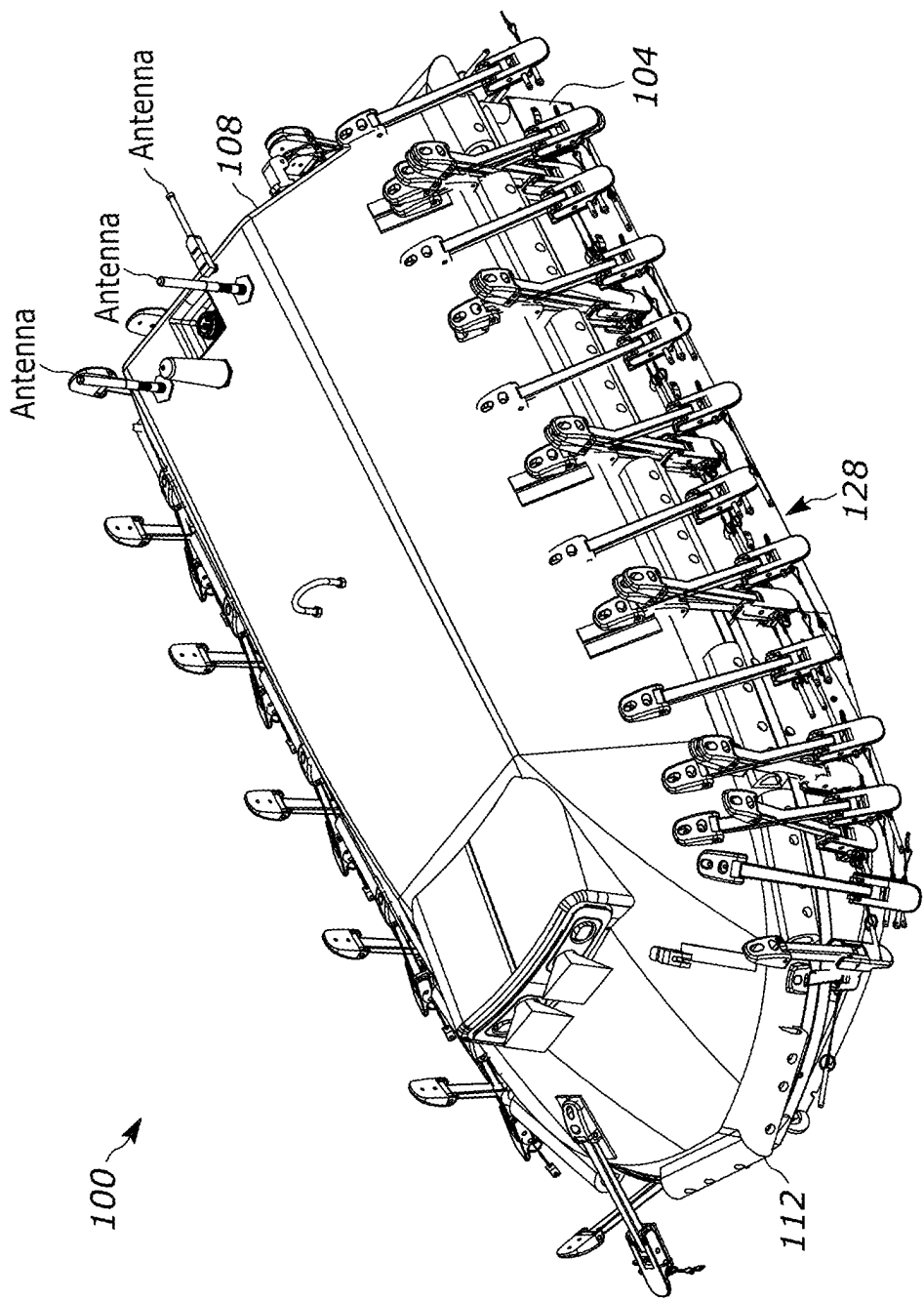


FIG. 1K

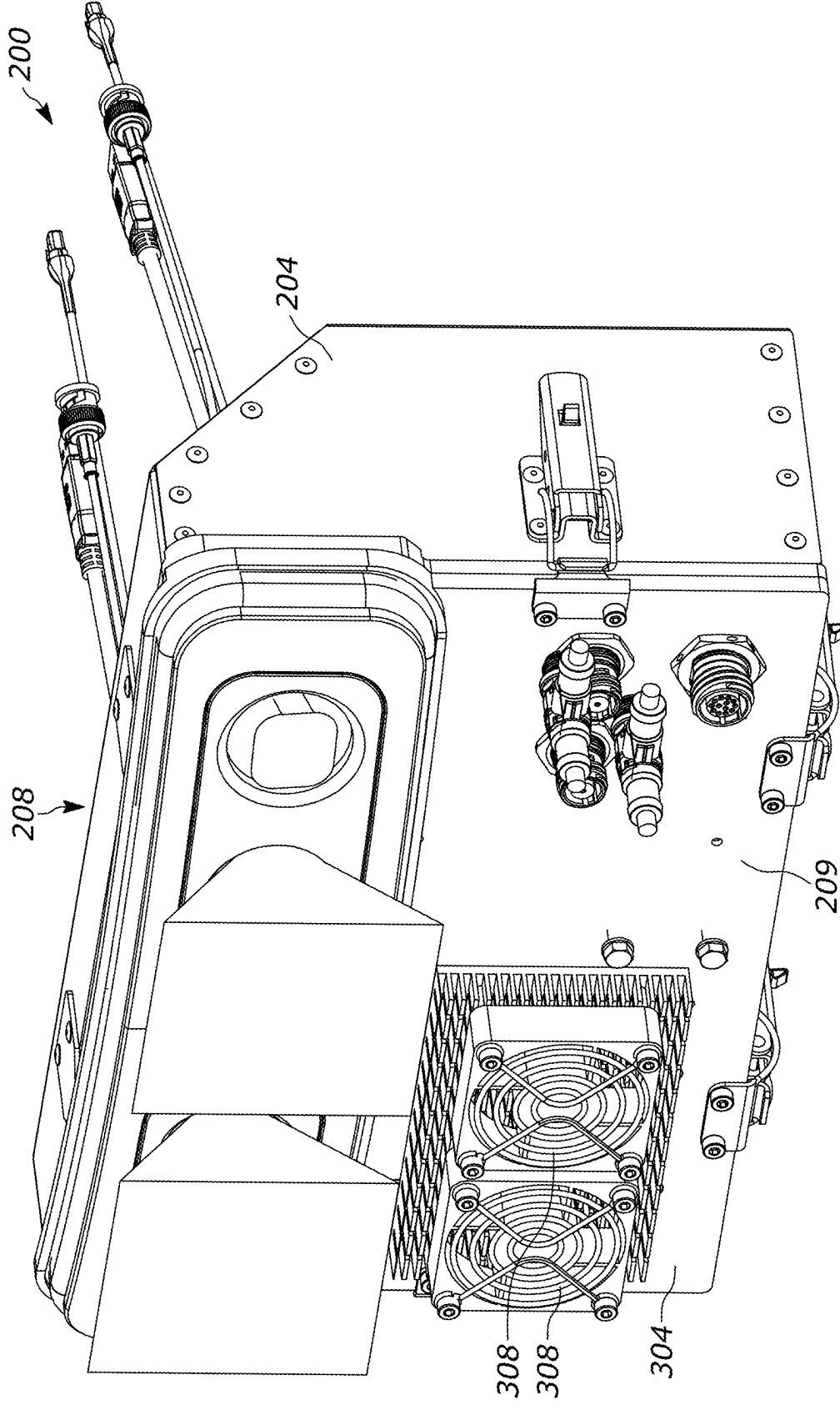


FIG. 2A

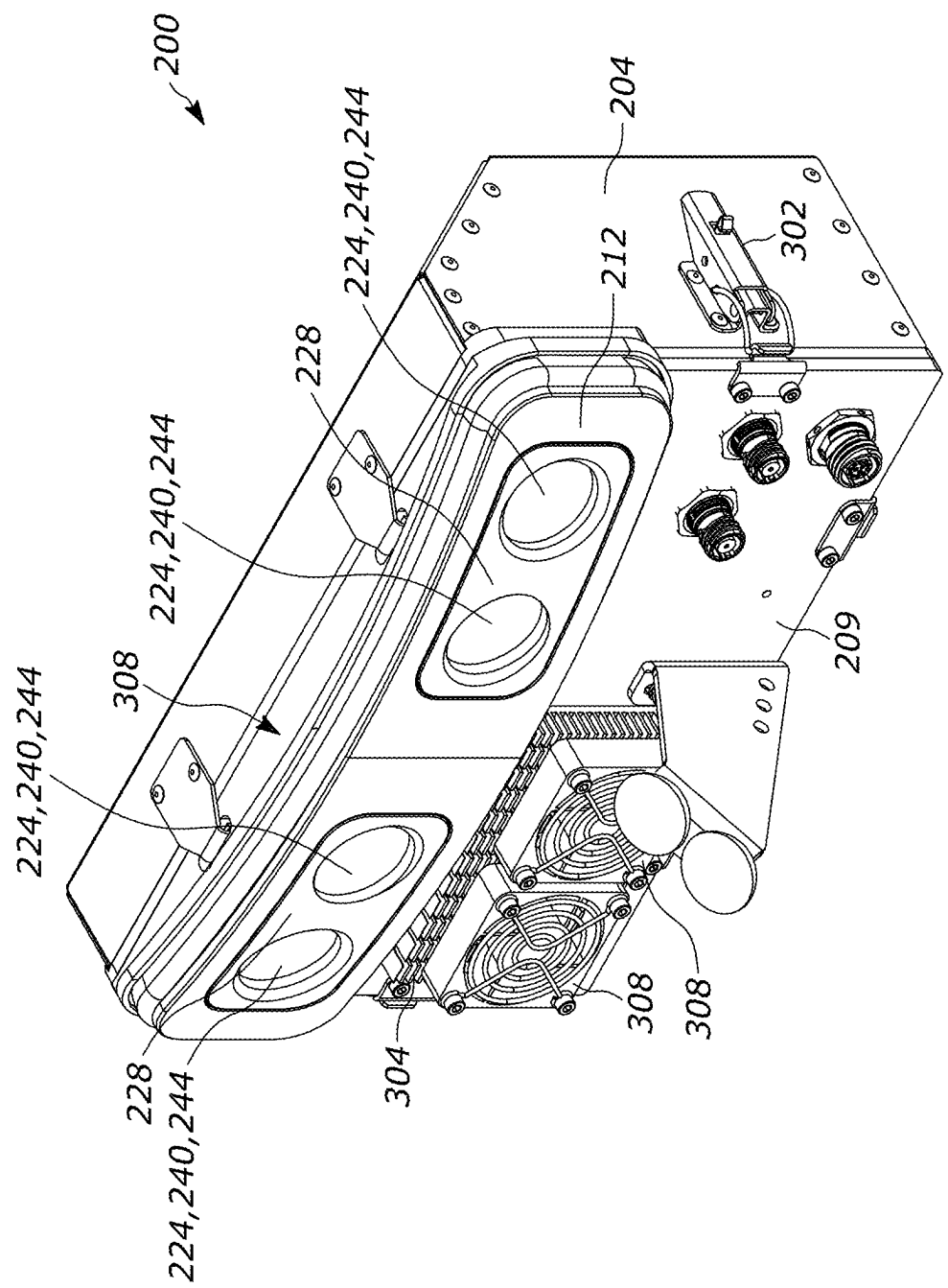


FIG. 2B

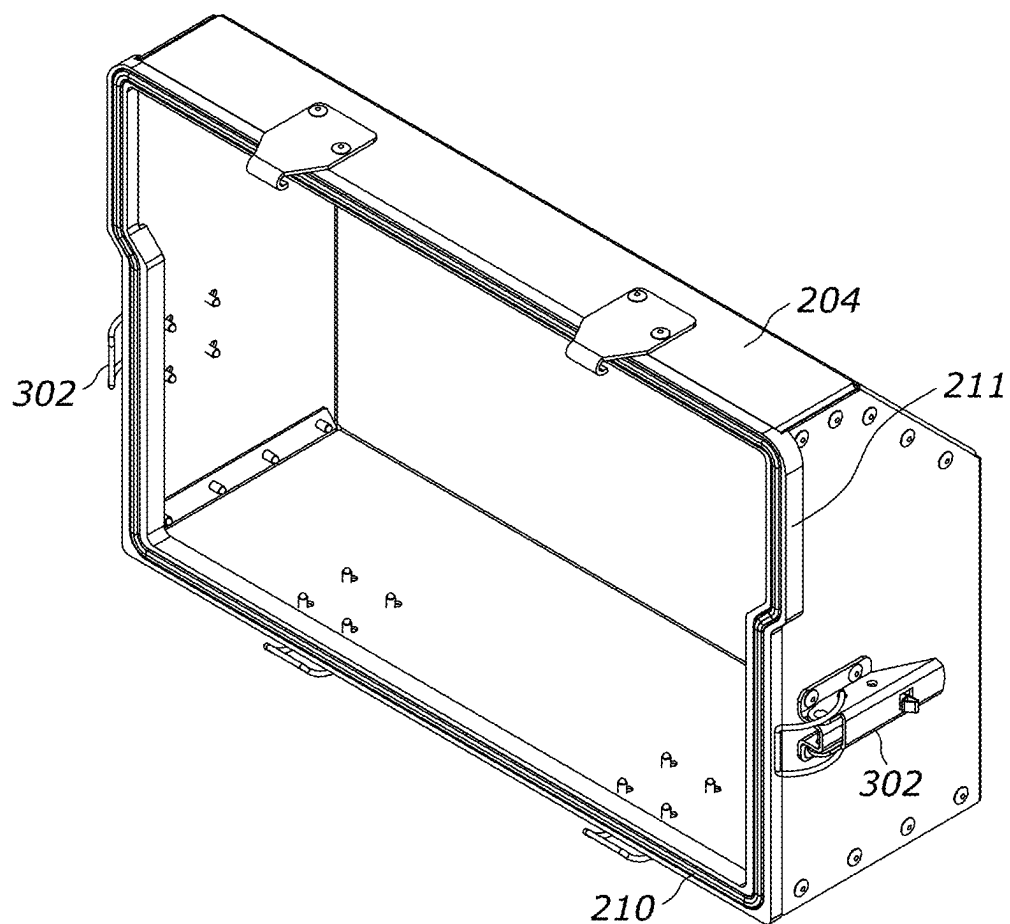


FIG. 2C

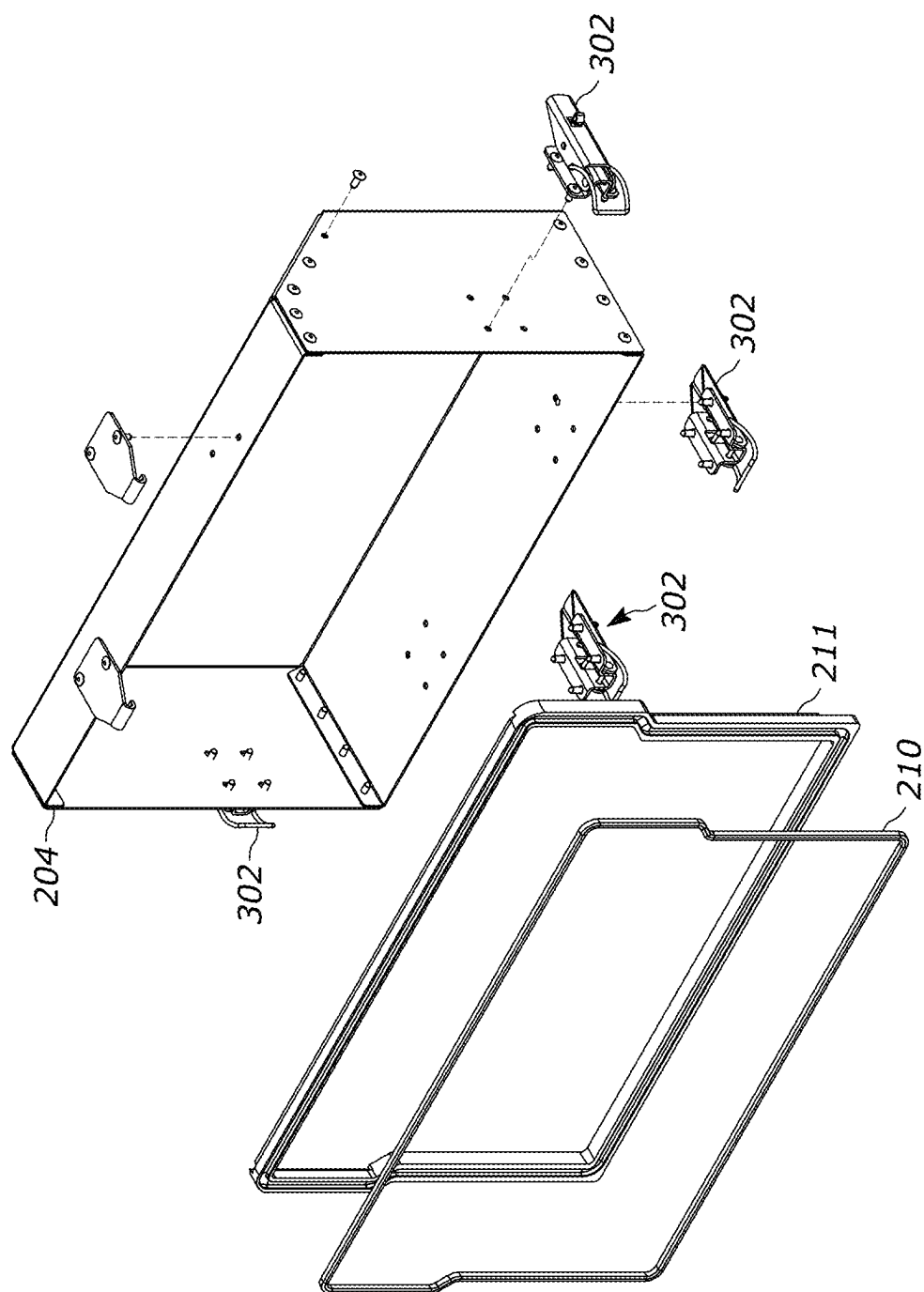


FIG. 2D

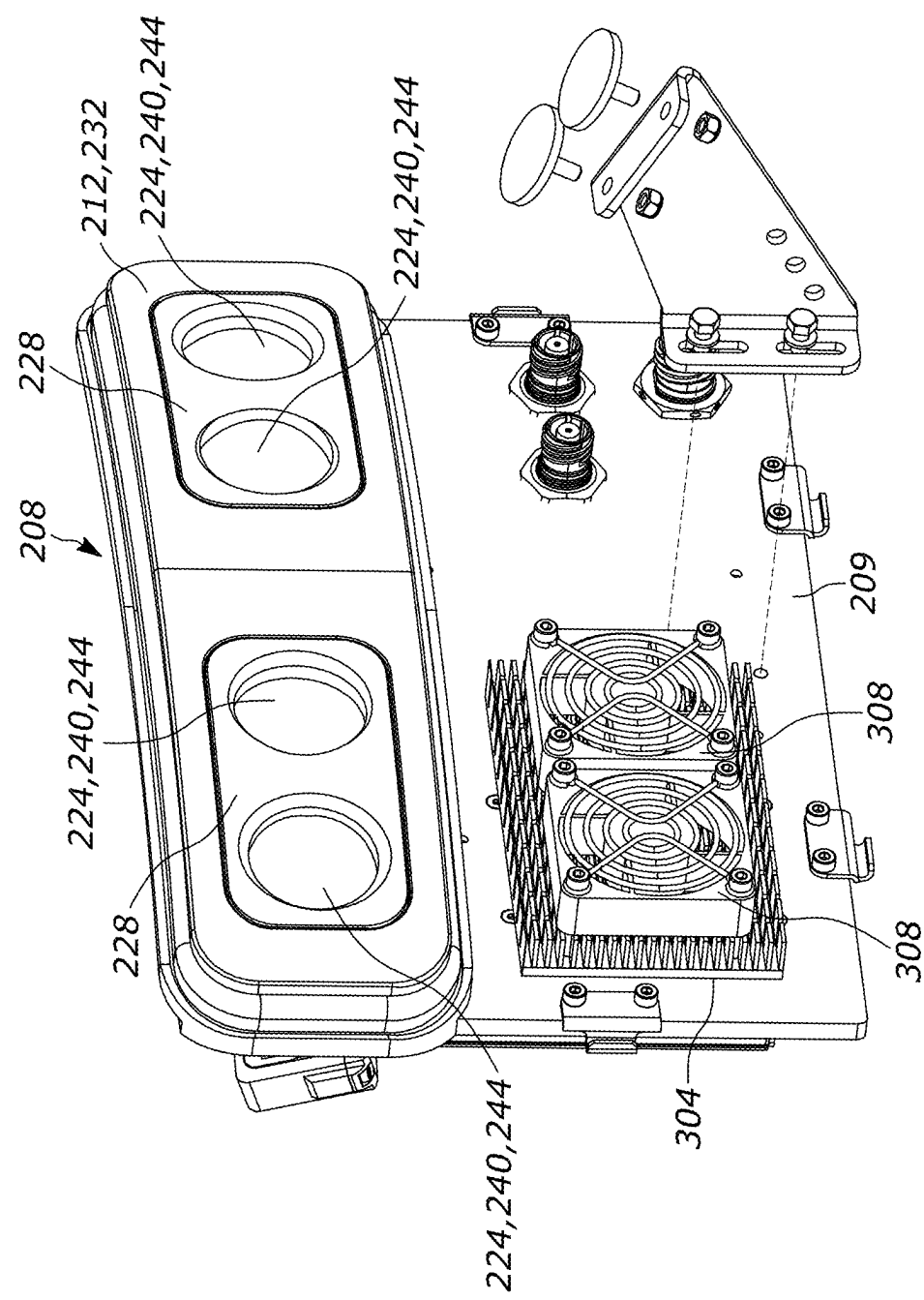


FIG. 2E

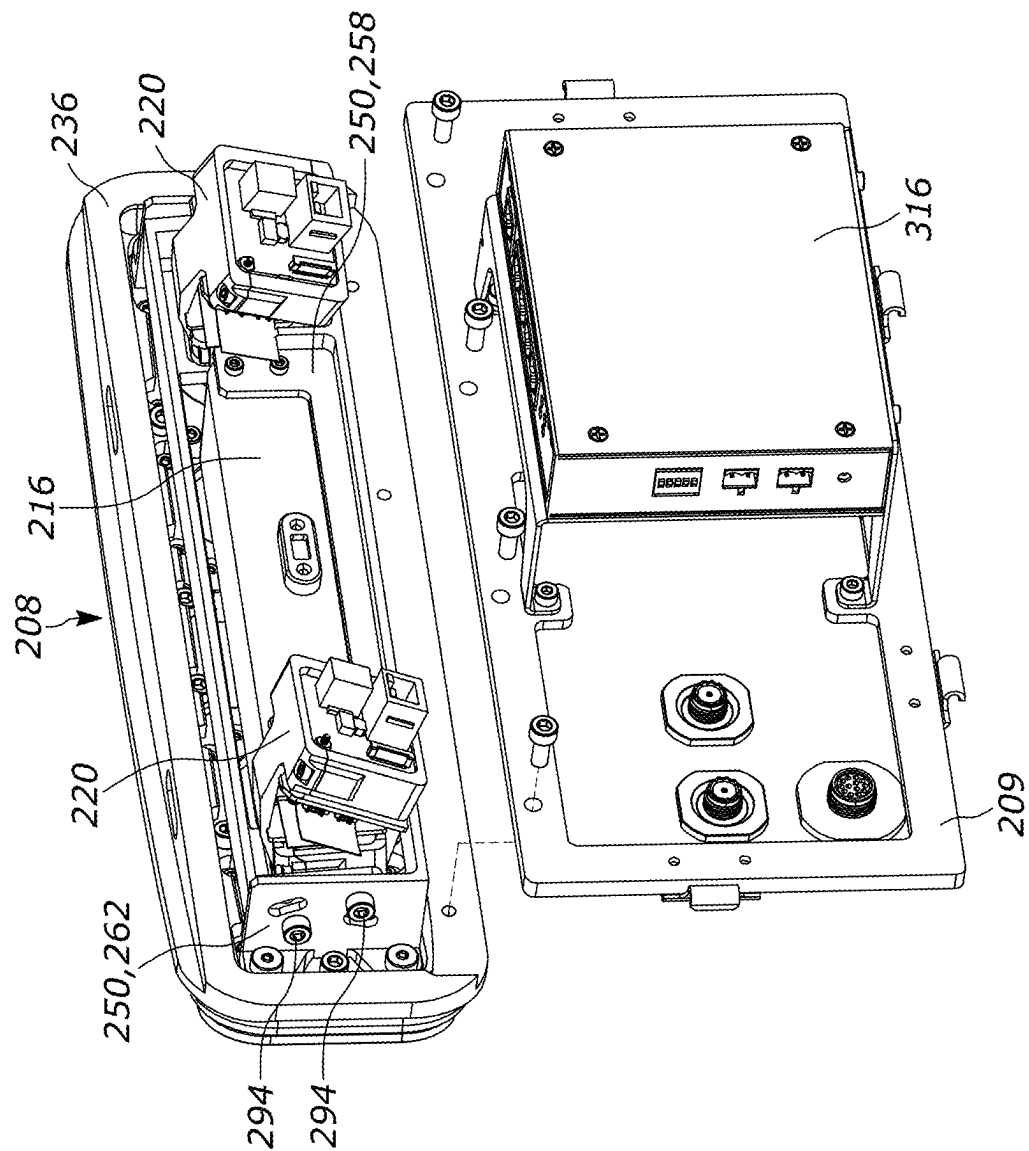


FIG. 2F

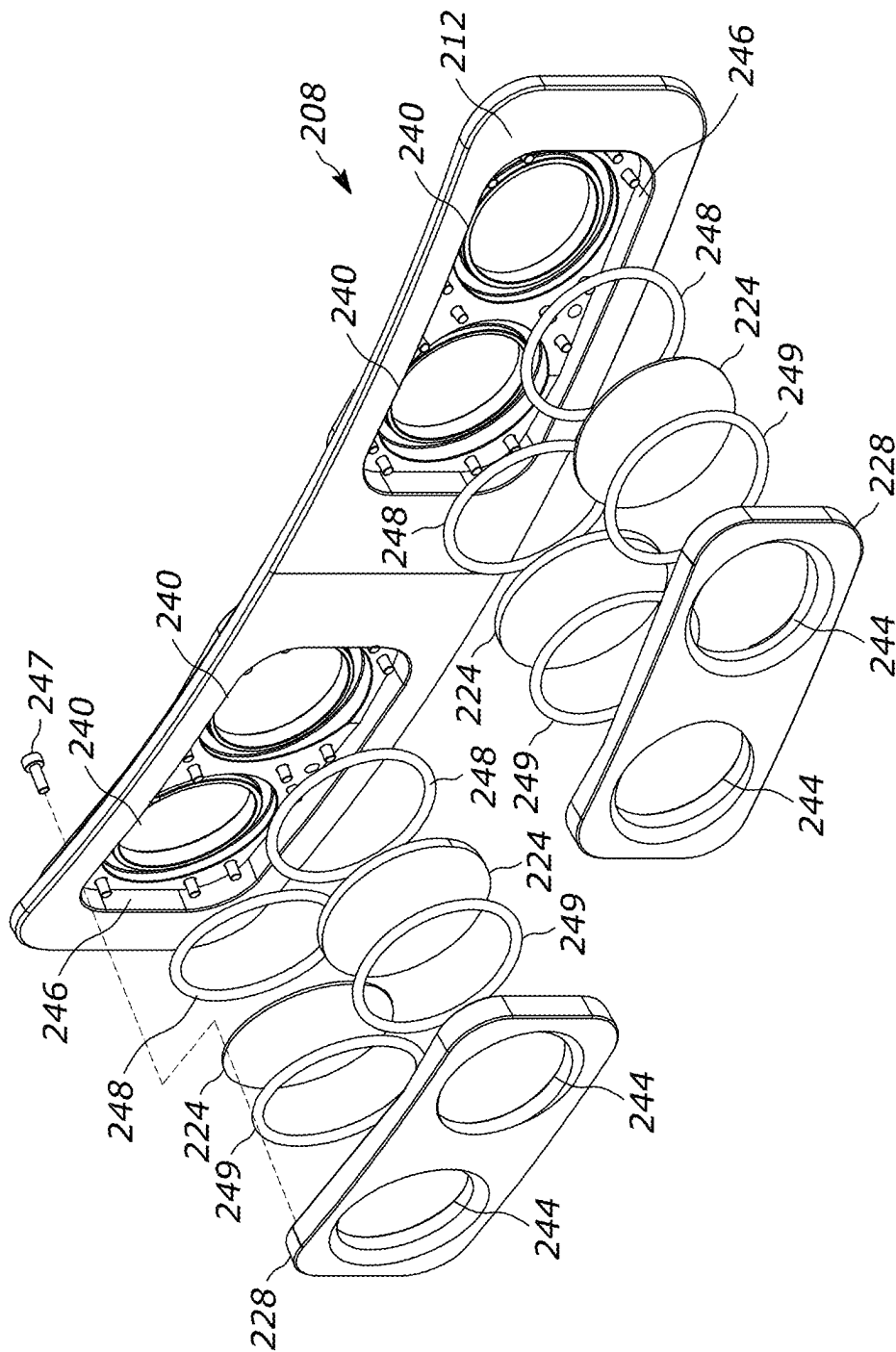


FIG. 2G

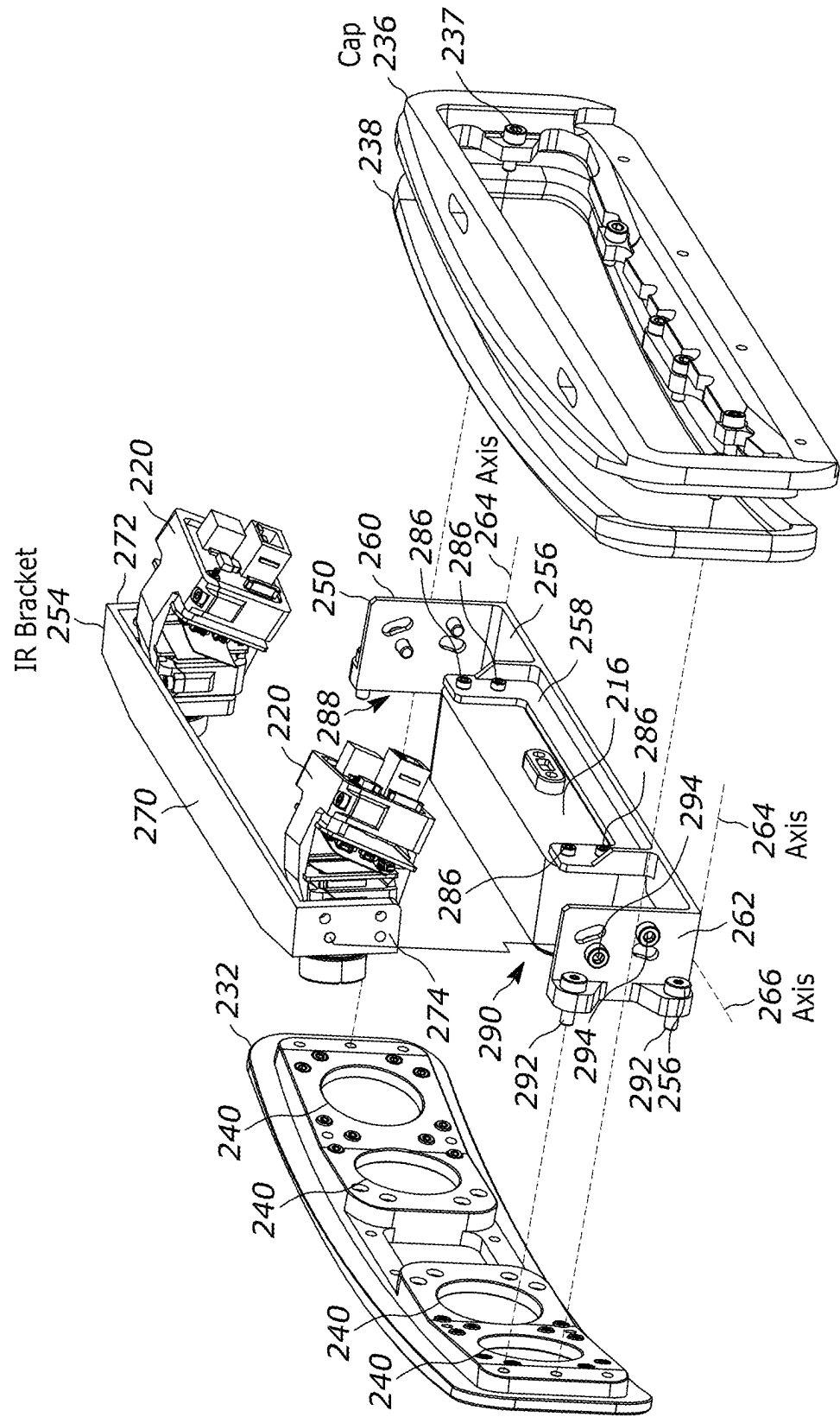


FIG. 2H

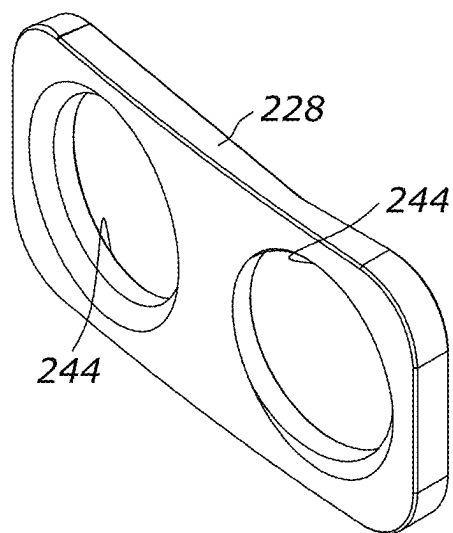


FIG. 2I

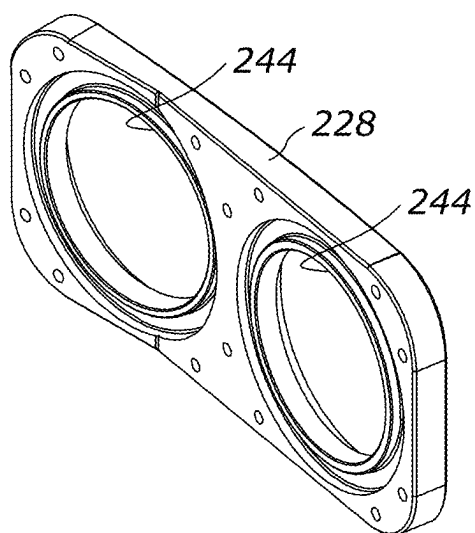


FIG. 2J

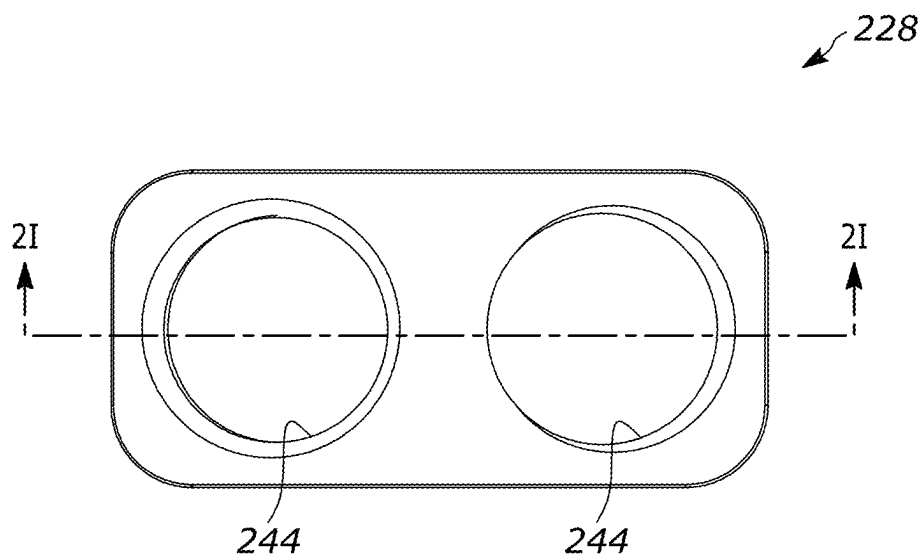


FIG. 2K

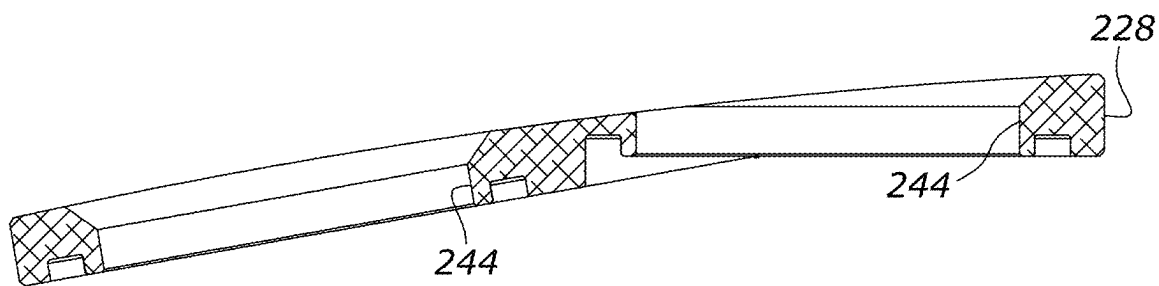


FIG. 2L

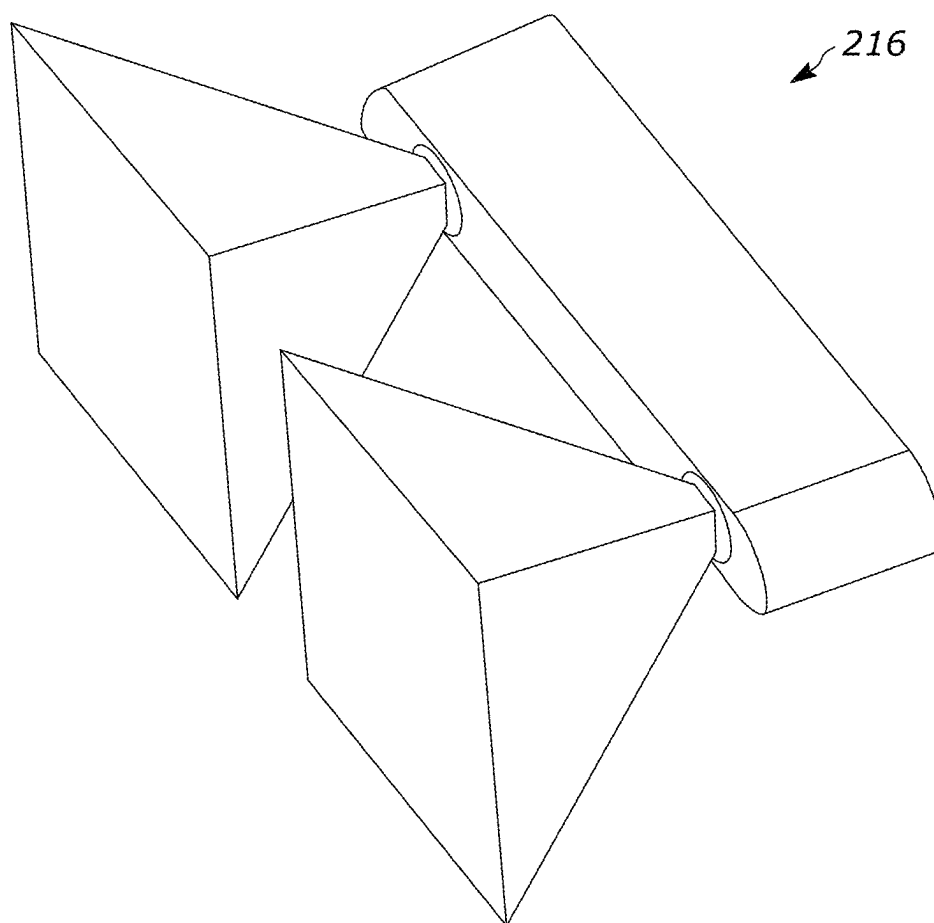


FIG. 2M

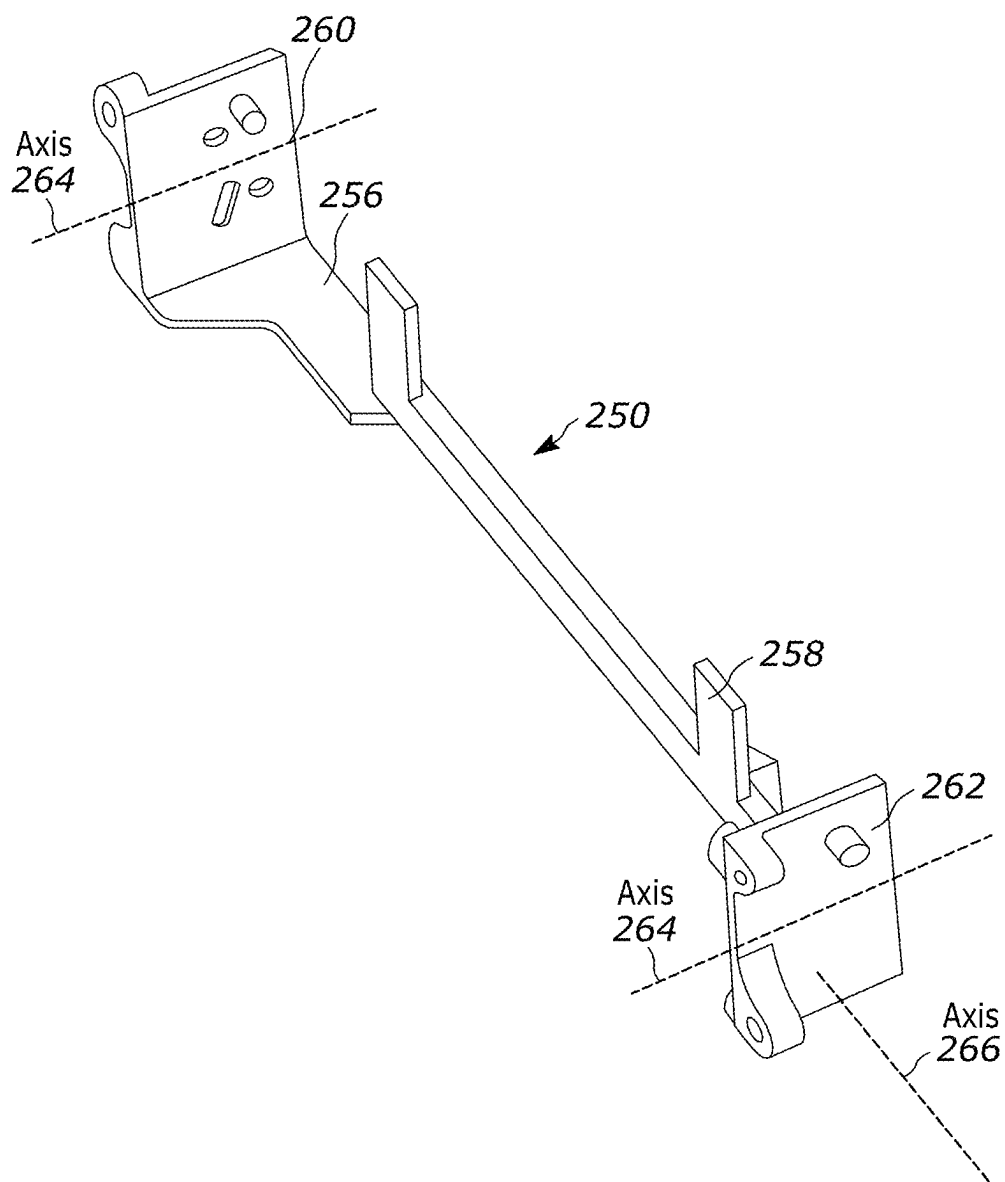


FIG. 2N

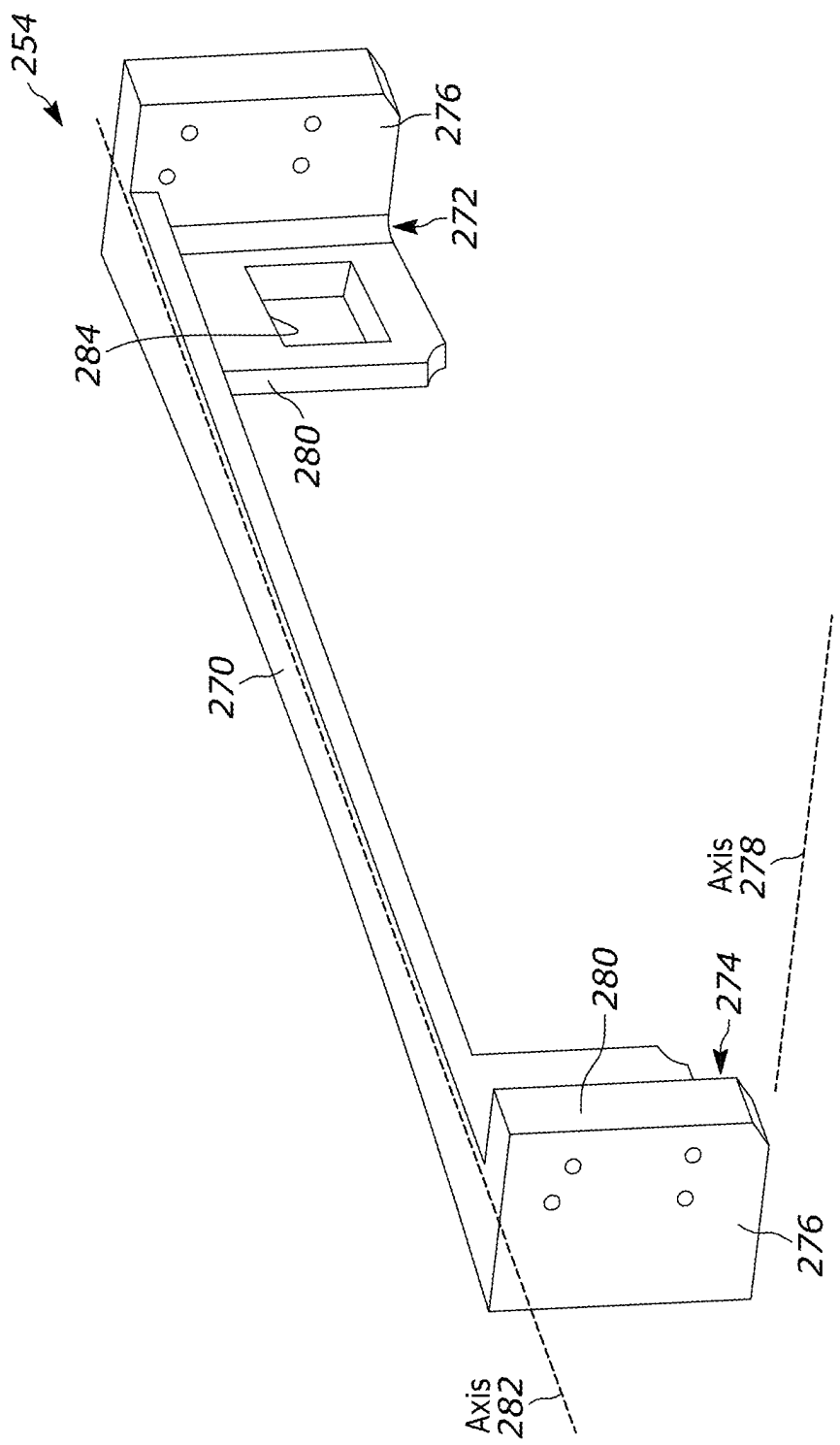


FIG. 20

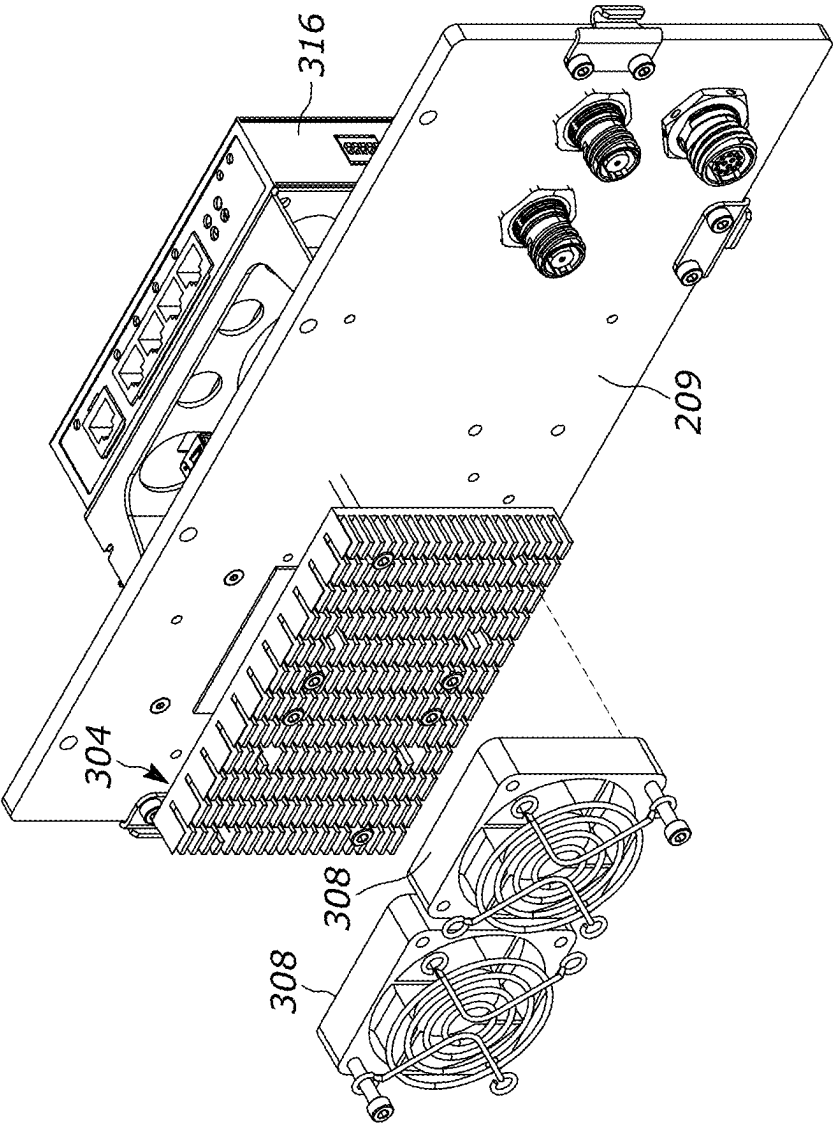


FIG. 2P

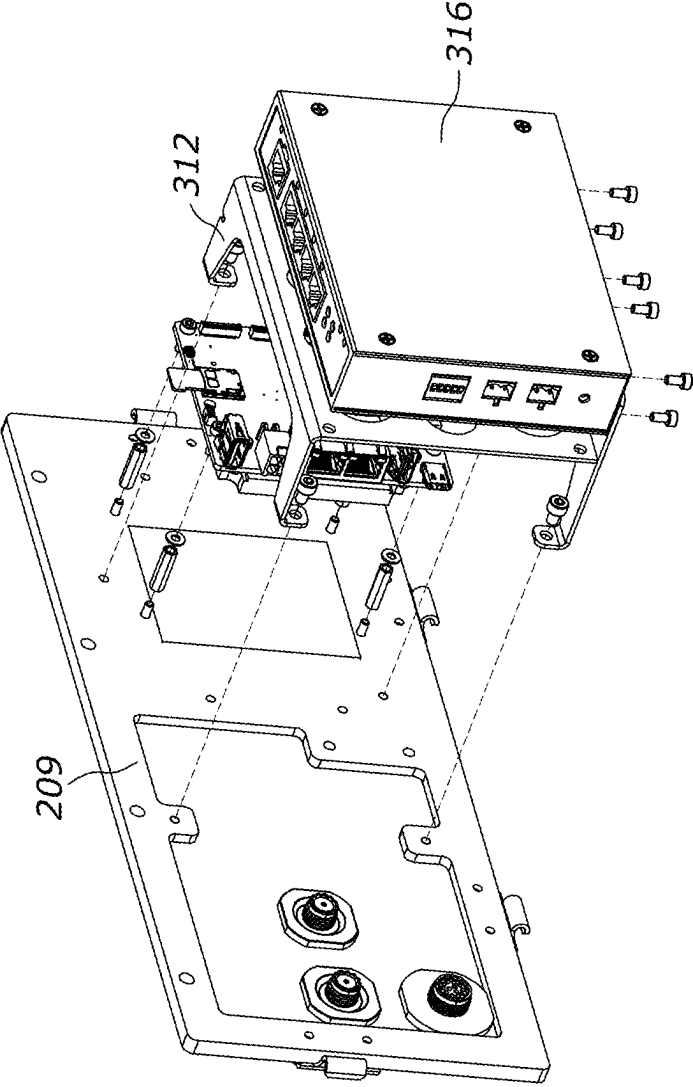


FIG. 2Q

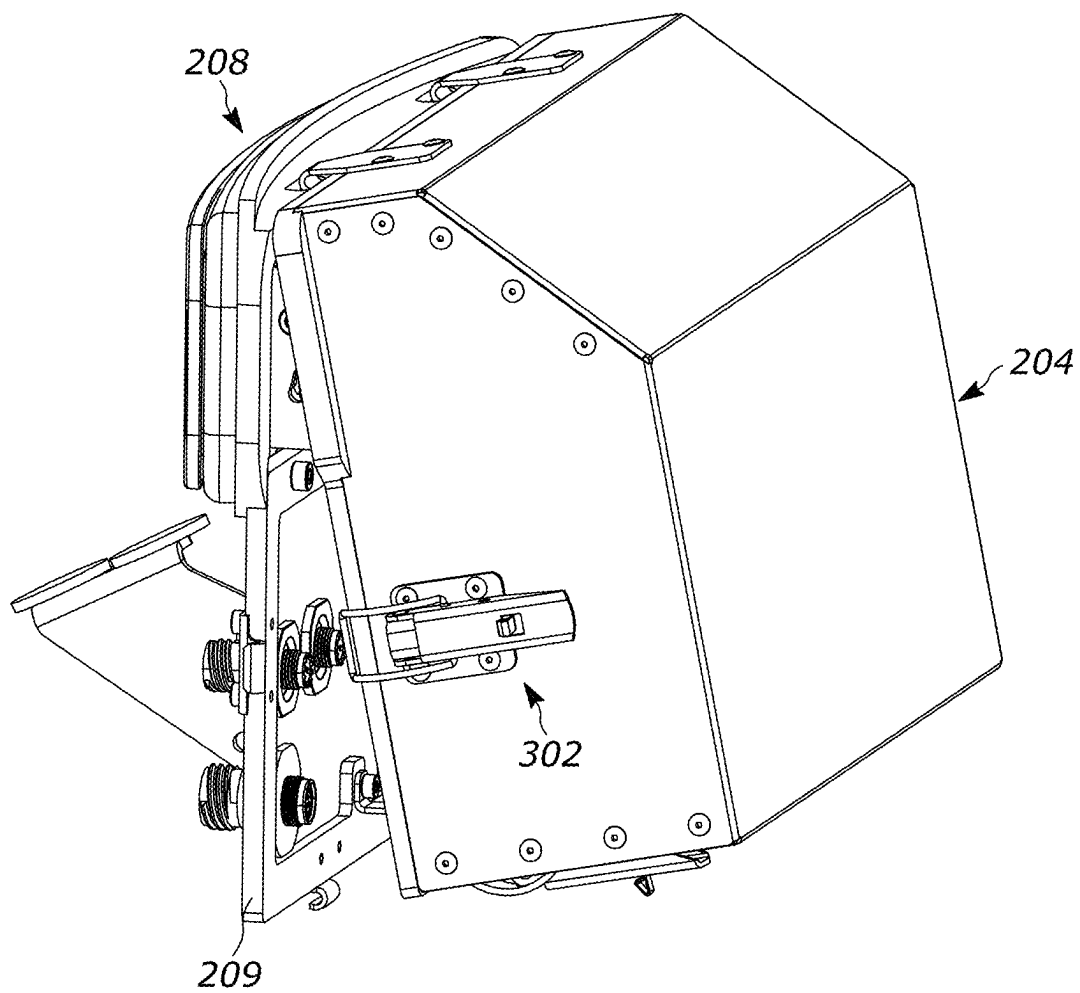


FIG. 2R

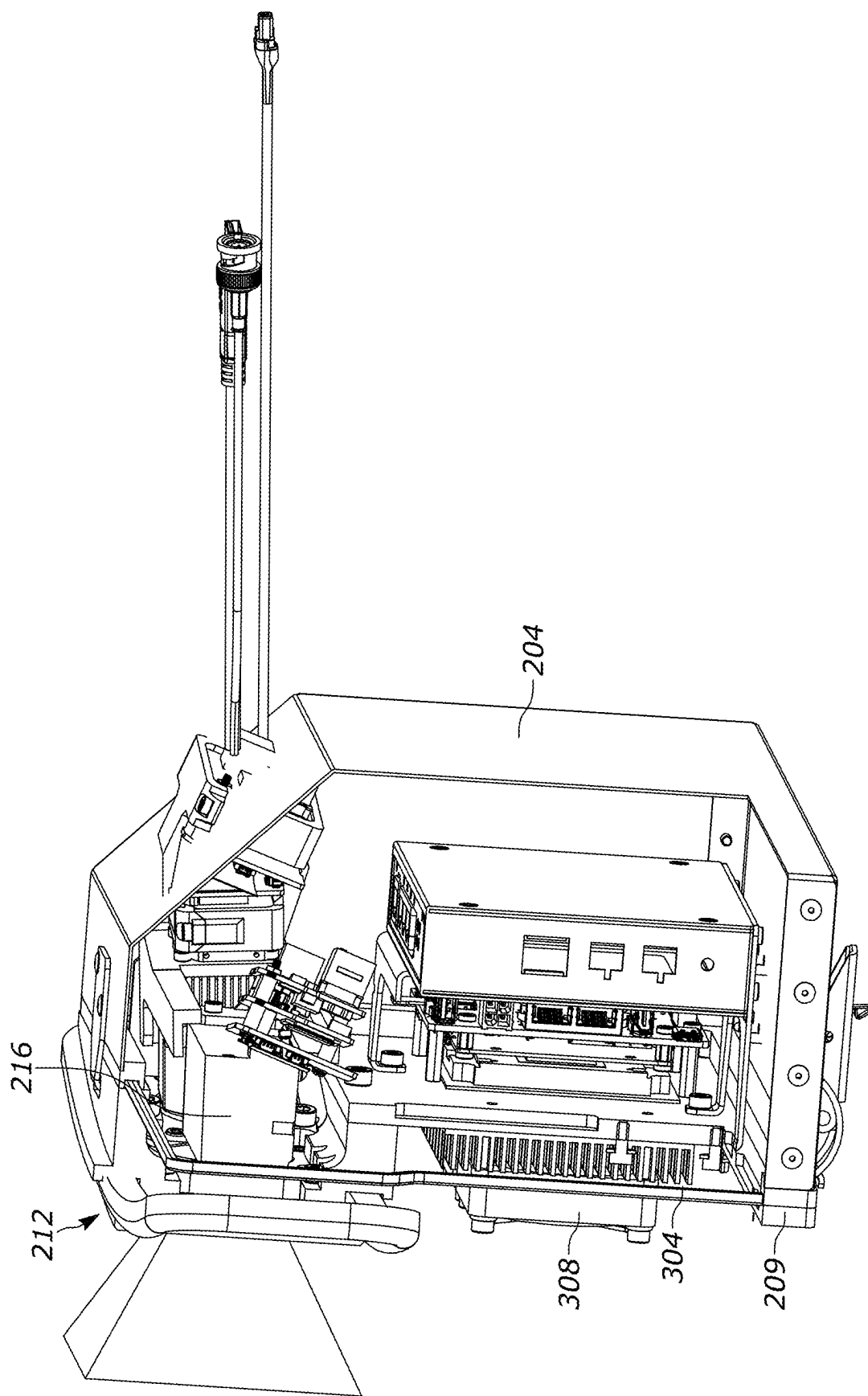


FIG. 2S

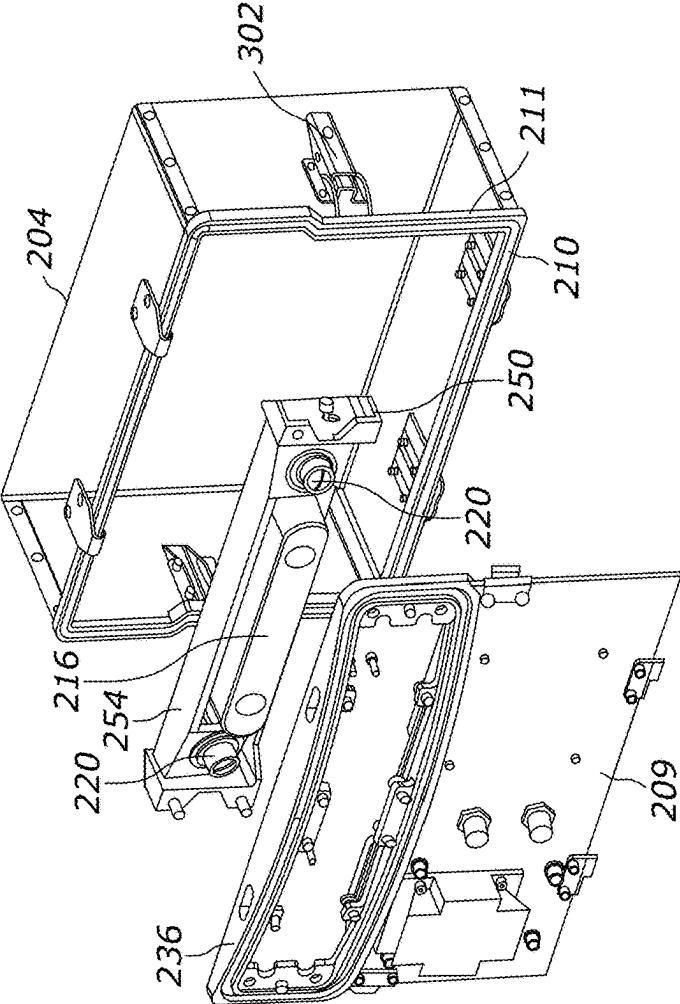
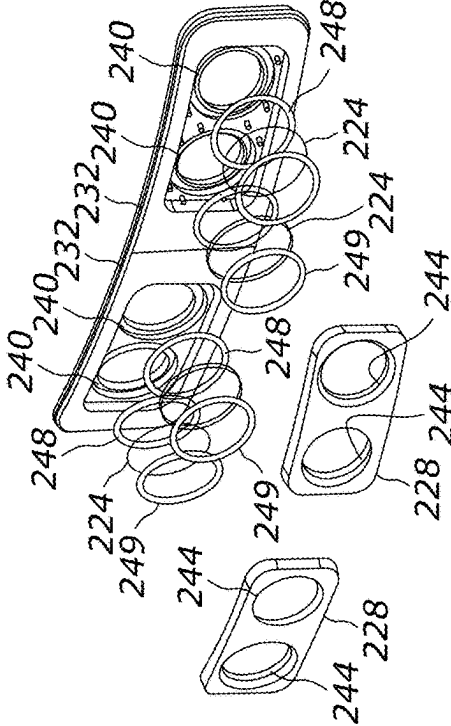


FIG. 2T



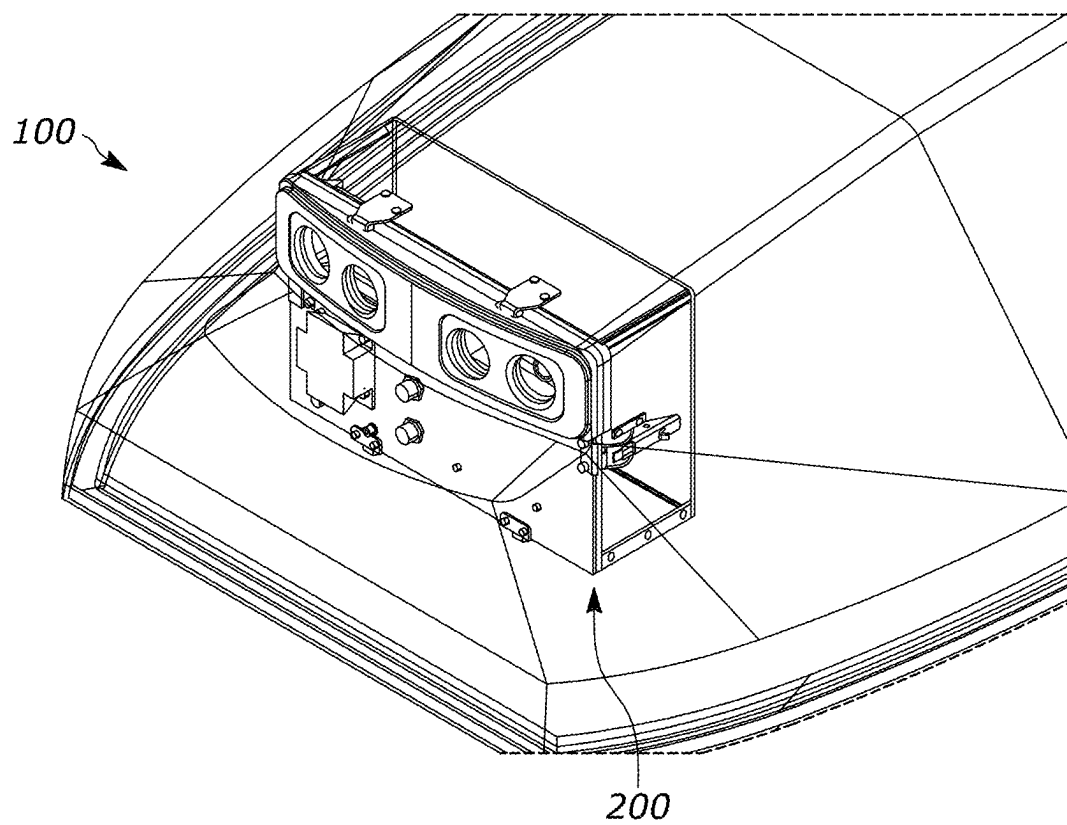


FIG. 2U

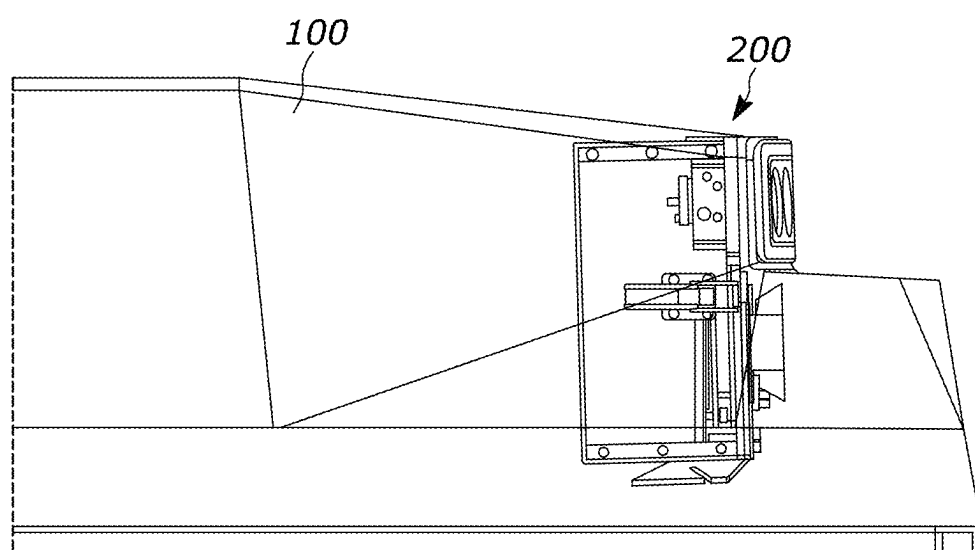


FIG. 2V

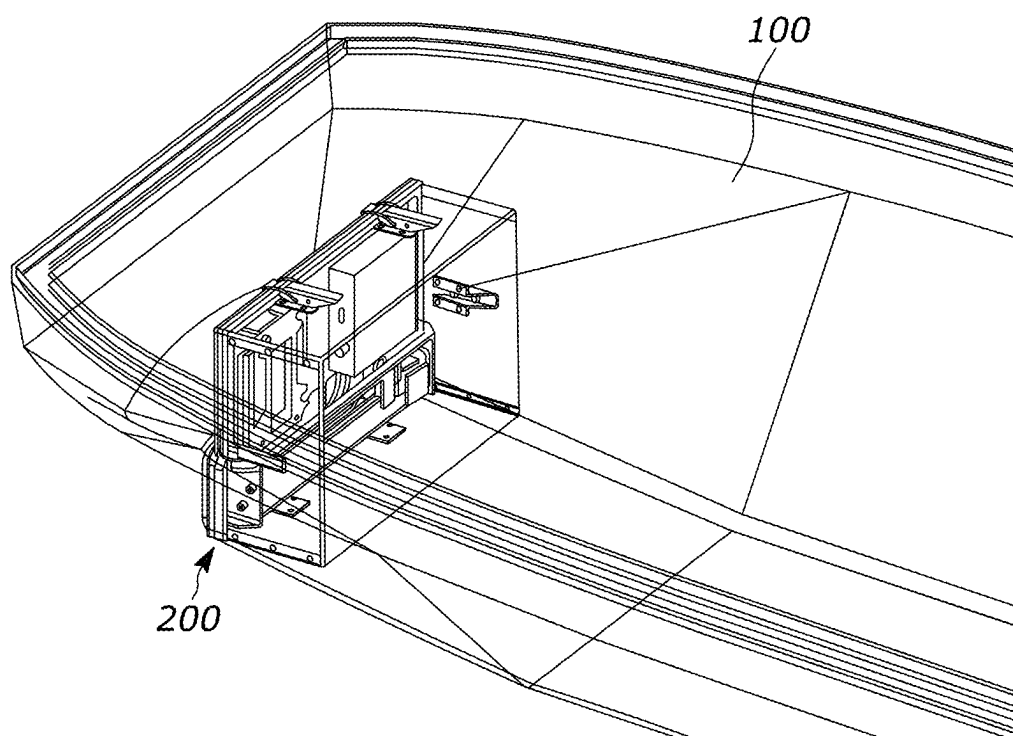
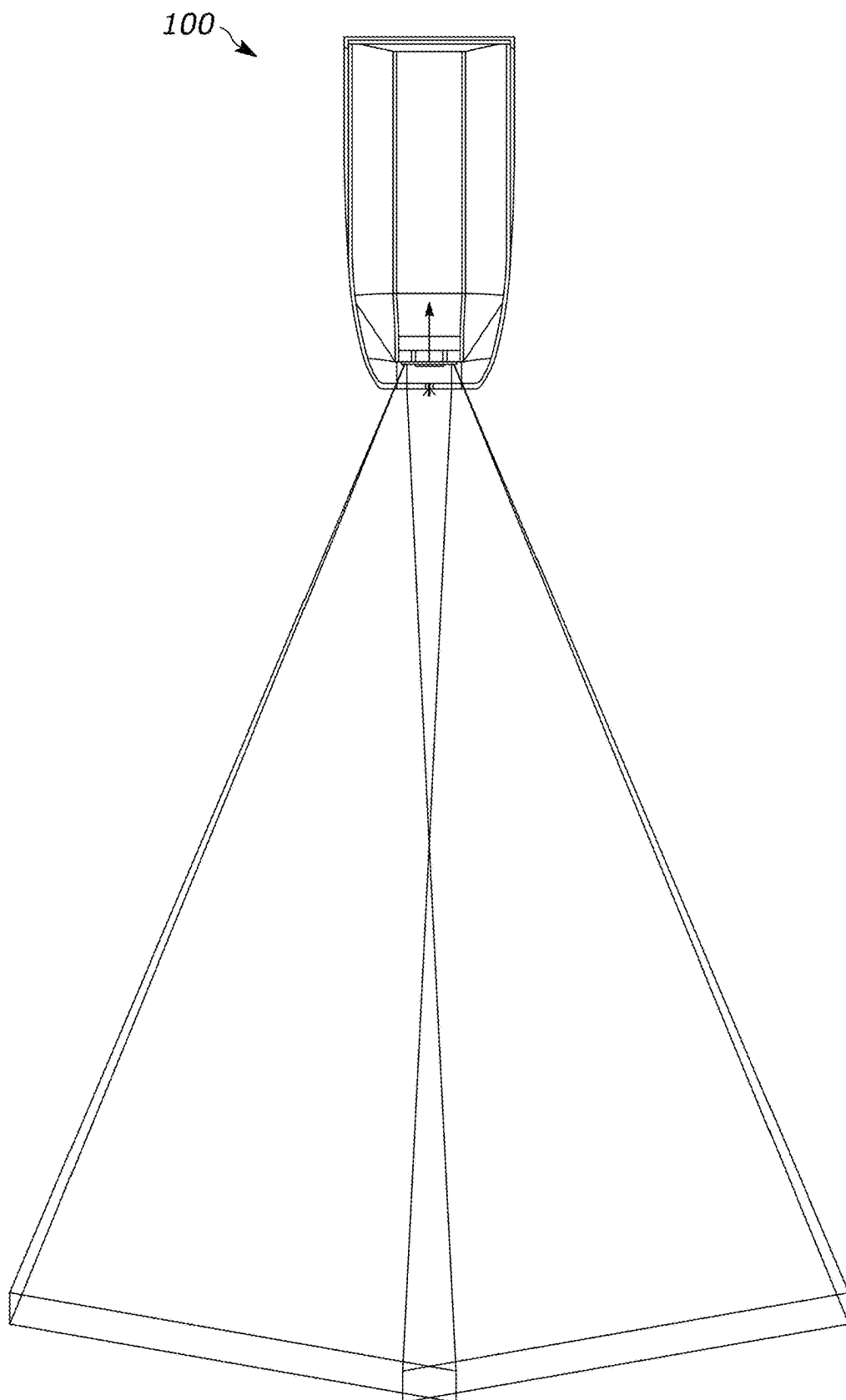


FIG. 2W



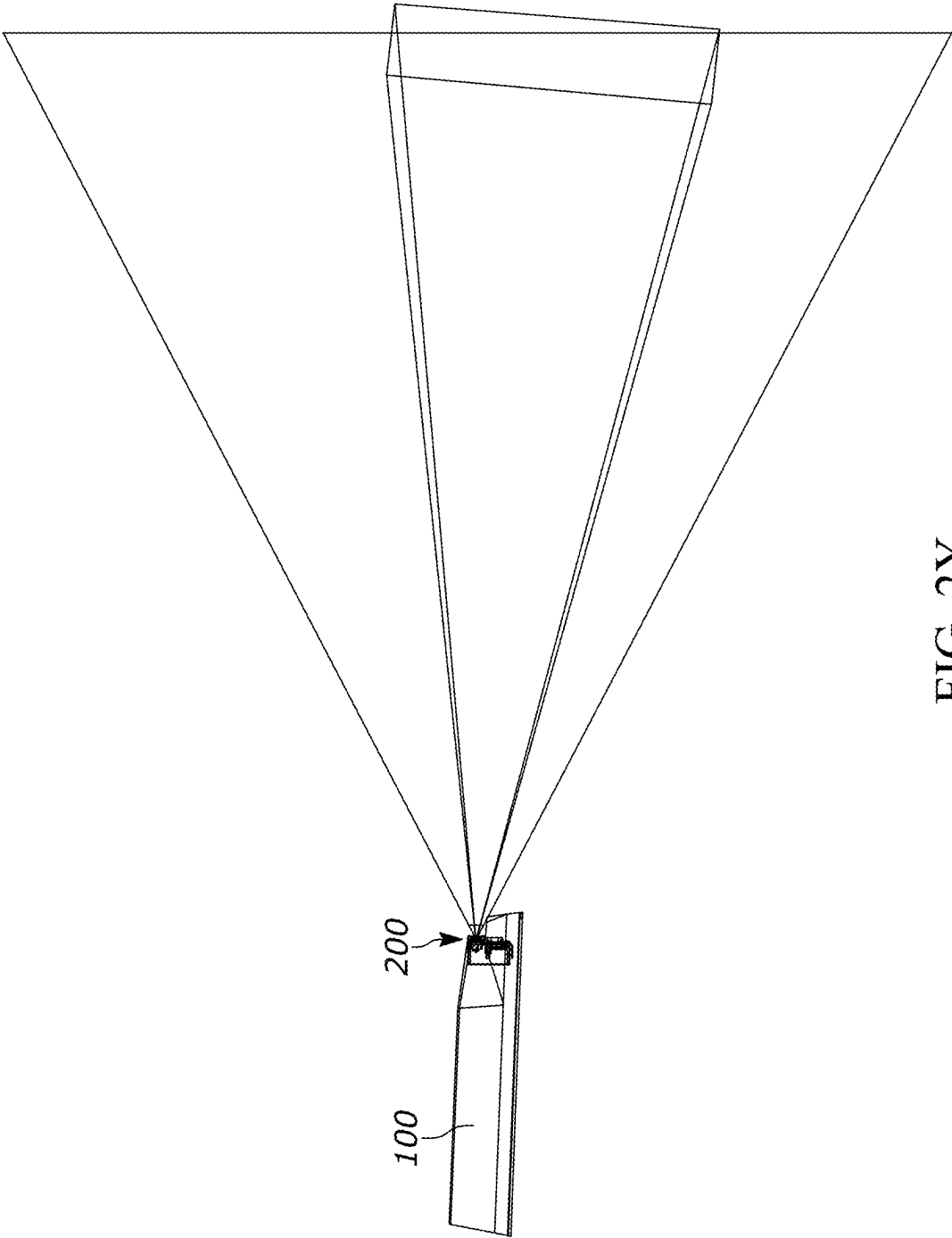


FIG. 2Y

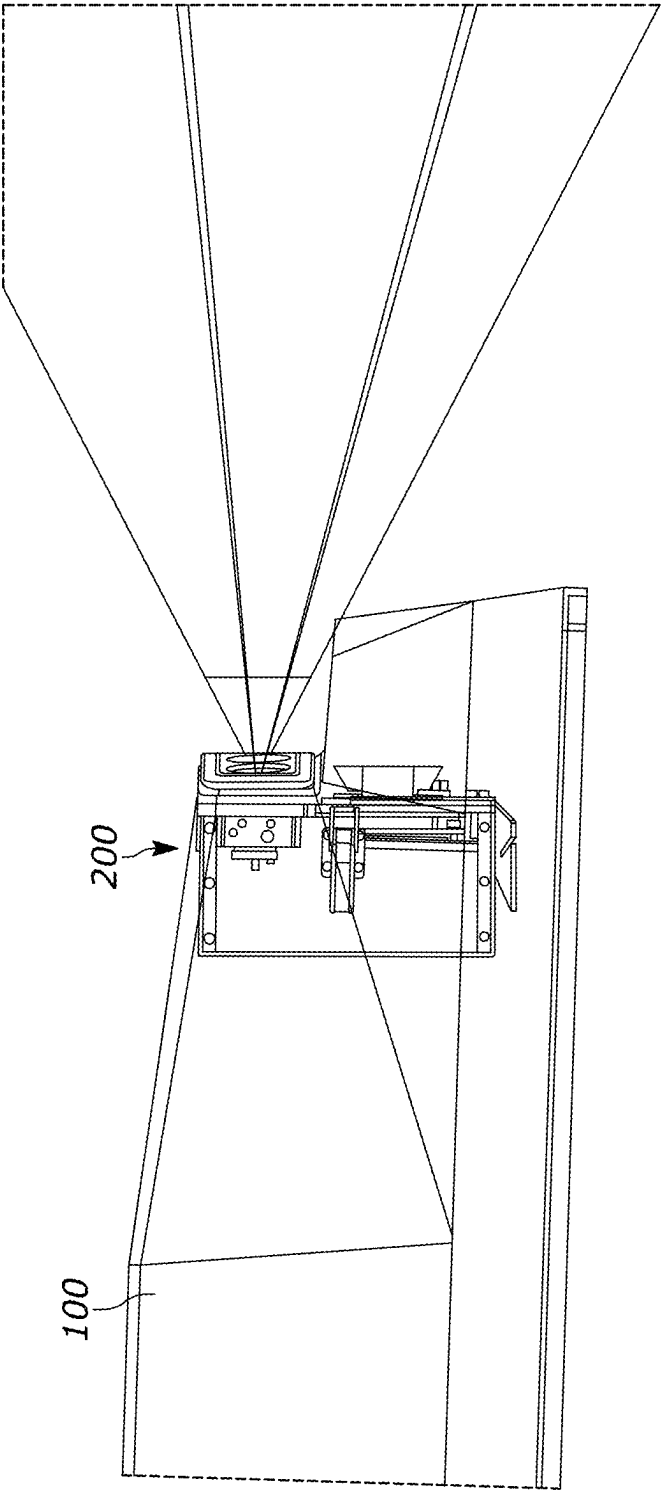


FIG. 2Z

VISION SYSTEM FOR MARITIME VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/561,301, titled “Vision System for Maritime Vehicle,” and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure generally relates to maritime vehicles and more specifically to a vision system for a maritime vehicle.

BACKGROUND OF THE DISCLOSURE

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

SUMMARY

[0004] A camera module for a maritime vehicle. The camera module includes a frame comprising first and second openings, first and second windows disposed in the frame immediately adjacent the first and second openings, respectively, a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings, a stereo camera disposed within the frame, the stereo camera disposed immediately adjacent the first opening, and an infra-red (IR) camera disposed within the frame, the IR camera disposed immediately adjacent the second opening.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The features of this invention which are believed to be novel are set forth with particularity in the appended claims. The invention may be best understood by reference to the following description taken in conjunction with the accompanying drawings, in which like reference numerals identify like elements in the several FIGS., in which:

[0006] FIG. 1A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;

[0007] FIG. 1B is a front view of the maritime vehicle of FIG. 1A;

[0008] FIG. 1C is a rear view of the maritime vehicle of FIG. 1A;

[0009] FIG. 1D is a bottom perspective view of the maritime vehicle of FIG. 1A;

[0010] FIG. 1E is a side view of the maritime vehicle of FIG. 1A;

[0011] FIG. 1F is similar to FIG. 1A, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0012] FIG. 1G is similar to FIG. 1F, but with additional components of the maritime vehicle removed for illustrative purposes;

[0013] FIG. 1H is a rear, perspective view of the maritime vehicle of FIG. 1F;

[0014] FIG. 1I is a first cross-sectional view taken along line I-I in FIG. 1A;

[0015] FIG. 1J is a second cross-sectional view taken along line J-J in FIG. 1A;

[0016] FIG. 1K is similar to FIG. 1A, but with the latching assembly of the maritime vehicle partially removed for illustrative purposes;

[0017] FIG. 2A is a front, perspective view of one example of a vision system that can be employed in the maritime vehicle of FIGS. 1A-1K;

[0018] FIG. 2B is another front, perspective view of the vision system of FIG. 2A;

[0019] FIG. 2C is a front, perspective view of a housing of the vision system of FIG. 2A;

[0020] FIG. 2D is an exploded view of the housing of FIG. 2C;

[0021] FIG. 2E is another front, perspective view of the vision system of FIG. 2A;

[0022] FIG. 2F is a rear, perspective view of a camera module and an electronics plate of the vision system of FIG. 2A;

[0023] FIG. 2G is a front, exploded view of the camera module;

[0024] FIG. 2H is a rear, exploded view of the camera module;

[0025] FIG. 2I is a front, perspective view of a retention plate of the camera module;

[0026] FIG. 2J is a rear view of FIG. 2I;

[0027] FIG. 2K is a top view of FIG. 2I;

[0028] FIG. 2L is a cross-sectional view taken along line K-K in FIG. 2K;

[0029] FIG. 2M is a perspective view of an electro-optical camera of the vision system of FIG. 2A;

[0030] FIG. 2N is a perspective view of a bracket for the electro-optical camera of FIG. 2M;

[0031] FIG. 2O is a perspective view of a bracket for infrared cameras of the vision system of FIG. 2A;

[0032] FIG. 2P is a front, perspective view of the electronics plate of the vision system of FIG. 2A;

[0033] FIG. 2Q is a rear, perspective view of the electronics plate;

[0034] FIG. 2R illustrates the electronics plate being coupled to the housing;

[0035] FIG. 2S is a cross-sectional view of the vision system of FIG. 2A;

[0036] FIG. 2T is an exploded view of the vision system of FIG. 2A;

[0037] FIG. 2U is a front, perspective view showing the orientation of the vision system of FIG. 2A when installed in the maritime vehicle of FIGS. 1A-1K;

[0038] FIG. 2V is a side view of FIG. 2U;

[0039] FIG. 2W is a rear, perspective view of FIG. 2U;

[0040] FIG. 2X is a top view of the field of vision of the vision system of FIG. 2A when in operation.

[0041] FIGS. 2Y and 2Z are side views of the field of vision of the vision system of FIG. 2A when in operation.

DETAILED DESCRIPTION

[0042] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but

can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0043] FIGS. 1A-1K illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vessel configured to autonomously traverse a body of water. The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** in this example is a mono-hull that has a front (or bow) **112**, a rear (or stern) **116**, two sides **120**, and a keel **124** coupled to another. The front **112**, the rear **116**, the sides **120**, and the keel **124** can be welded together or can be coupled to one another in a different manner. For example, the front **112**, the rear **116**, the sides **120**, and the keel **124** can be coupled together in the manner described in U.S. Provisional Application No. 63/561,282, titled “Systems and Approaches for Assembling a Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The hull **104** is configured such that the hull provides a continuous planning surface that allows the maritime vehicle **100** to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap **108** is coupled to the hull **104** to cover and/or conceal the components of the maritime vehicle **100** disposed in and carried by the hull **104** as the maritime vehicle **100** traverses the body of water.

[0044] In this example, the hull **104** and the cap **108** each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull **104** is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle **100** can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull. In this example, the cap **108** entirely covers the hull **104** (and the components therein). In other examples, however, the maritime vehicle **100** need not include the cap **108** or the cap **108** may only partially cover the hull **104** (and the components disposed therein).

[0045] In some examples, the cap **108** can be removably coupled to the hull **104** via a locking system. For example, as illustrated in FIGS. 1A-1K, the locking system can take the form of a plurality of latch mechanisms **128** disposed around at least a portion (if not the entirety) of a perimeter of the maritime vehicle **100**. Thus, the cap **108** can be removed to allow access to the interior of the hull **104**. In other examples, however, the cap **108** can be permanently coupled to the hull **104** to permanently conceal the components within the maritime vehicle **100**.

[0046] The maritime vehicle **100** also includes a plurality of bulkheads **132** arranged within the hull **104**. The bulk-

heads **132** divide the maritime vehicle **100** into a plurality of different compartments for receiving and retaining different components in the maritime vehicle **100**.

[0047] The maritime vehicle **100** also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle **100** as well as data about the environment surrounding the maritime vehicle **100** (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle **100**. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle **100**, the pressure of the water in which the maritime vehicle **100** is disposed, the pressure within the maritime vehicle **100**), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle **100**, a temperature of ambient air external to the maritime vehicle **100**, a temperature of water in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0048] The maritime vehicle **100** also includes a power system that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system generally includes a thrust system and one or more power sources configured to power the thrust system (and the other components within the maritime vehicle **100**). The thrust system is generally configured to propel the maritime vehicle **100** in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system such as the jet pump assembly described in U.S. Provisional Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. 1A-1K includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** illustrated in FIGS. 1A-1K also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The maritime vehicle **100** generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the

maritime vehicle **100**. For example, the maritime vehicle **100** can include the cooling system described in U.S. Provisional Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein.

[0049] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0050] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and resurfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). The vision system is generally configured to capture, process, and analyze images obtained by the one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment).

[0051] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae disposed on an exterior of the cap **108** as well as a plurality of antennae disposed in the hull **104**.

[0052] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more thrust controllers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0053] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, and/or one or more brain boards.

[0054] As discussed above, the maritime vehicle **100** can include a vision system. FIGS. 2A-2Z illustrate one example of a vision system **200** that can be employed in the maritime vehicle **100**. In this example, the vision system **200** is a stereoscopic vision unit that includes two independent stereoscopic cameras. In this example, the vision system **200** is fully assembled and calibrated before being mounted to a front of the maritime vehicle **100**. The vision system **200** can, for example, be sealingly and securely mounted to a front of the hull cap **108** (see, for example, FIGS. 1A and 1K). More particularly, the vision system **200** includes an external element or interface that is bonded with the front of the hull cap **108** as well as one or more seals (e.g., a double o-ring barrel seal) that sealingly couple the external element or interface and the rest of the vision system **200**. In this example, the vision system **200** is also removable from the maritime vehicle **100** and can be replaced with another vision system **200** (or a different vision system). In other examples, the vision system **200** can be fully assembled and calibrated while or after being mounted to the front of the maritime vehicle **100**. In any event, the vision system **200**, so mounted, is optimally positioned to capture, process, and analyze data about the environment surrounding the maritime vehicle **100**. Broadly speaking, the vision system **200** is configured to passively detect or sense the presence and/or absence of an object (and/or of any objects, for that matter).

within the FOV of the vision system **200** and enable accurate depth (i.e., distance) estimates of such objects using stereoscopic techniques.

[0055] The vision system **200** generally includes a housing **204**, a camera module **208** coupled to and carried by the housing **204**, and an electronics plate **209** coupled to and carried by the housing **204**. The housing **204** is configured to be mounted to the maritime vehicle **100**, and, more particularly, to the front of the hull cap **108** (see FIGS. **1A** and **1K**). For example, the housing **204** can include the external element or interface that is bonded with the front of the hull cap **108**. The housing **204** is preferably made of fiberglass but can be made of another strong material such that the housing **204** protects the camera module **208** when the maritime vehicle **100** experiences significant shock (e.g., shock values up to 20G), traverses the body of water at high speeds, or is used in dangerous conditions. In some examples, the housing **204** can be purged with nitrogen or dry air for improved performance. Preferably, and as best illustrated in FIGS. **2C**, **2D**, and **2T**, the vision system **200** also includes a first sealing element (e.g., a gasket) **210** and a mount **211** for mounting the first sealing element **210** to the housing **204**. In this example, the mount **211** has a substantially rectangular shape and is coupled (e.g., fixedly or removably coupled) to an outer perimeter edge of the housing **204**, and the first sealing element **210** has a similar shape as the mount **211** and is removably disposed in a channel formed in the mount **211**. When the first sealing element **210** is mounted to the housing **204** via the mount **211**, the first sealing element **210** surrounds the interior of the housing **204**, such that the first sealing element **210** is arranged to sealingly engage the camera module **208** and the electronics plate **209** and to effect a seal between the housing **204** and the camera module **208** and the electronics plate **209** when the camera module **208** and the electronics plate **209** are coupled to the housing **204**. In other words, the first sealing element **210** serves to seal the electrical components within an interior of the housing **204** when the camera module **208** and the electronics plate **209** are coupled to the housing **204**.

[0056] The camera module **208** generally includes a frame **212**, one or more electrical-optical (“EO”) cameras **216**, one or more infrared (“IR”) cameras **220**, a plurality of windows (or lenses) **224**, and a plurality of window retention plates **228** for the windows **224**. In this example, the frame **212** is defined by a face plate **232** and a cap **236** coupled to the face plate **232** via a plurality of fasteners **237** (one of which is illustrated in FIG. **2H**) and via adhesive (e.g., applied to the rear surface of the face plate **232**). In other examples, the face plate **232** and the cap **236** can be coupled together in a different manner. Preferably, the frame **212** also includes a second sealing element **238** (e.g., a gasket) secured to the cap **236** so as to be disposed between the face plate **232** and the cap **236**. In this example, the second sealing element **238** is secured in a groove formed in the front surface of the cap **236**. The frame **212** also includes a plurality of openings **240** formed in the face plate **232** and sized to receive the windows **224**. In this example, the frame **212** includes four openings **240** divided into two pairs of openings. In other examples, however, the frame **212** can include more or less openings **240**. Moreover, while not illustrated herein, it will be appreciated that the camera module **208** also generally includes a plurality of covers configured to selectively cover the plurality of windows **224**, respectively. The plurality of

covers may also cover the plurality of apertures **244** formed in the window retention plates **228**.

[0057] In this example, the camera module **208** includes a single electrical-optical camera **216** that preferably takes the form of a stereo camera with dual EO image sensors. In this example, the camera module **208** includes two infrared cameras **220**. Each of the infrared cameras **220** preferably takes the form of a stereovision IR camera, which may utilize one or more of short wave IR (SWIR), mid-wave IR (MWIR) (cooled or uncooled), and/or long-wave IR (LWIR) (cooled or uncooled) sensors. When the camera module **208** includes pairs of EO/IR cameras **216**, **220**, each camera may be configured to capture similar electromagnetic radiation across a similar FOV, and may be separated (e.g., fixedly separated) by a baseline distance.

[0058] In this example, because the frame **212** includes four openings **240**, the camera module **208** includes four windows **224**. In this example, each window of the plurality of windows **224** is flat, and each window of the plurality of windows **224** is oriented planar to the EO camera **216** and the IR cameras **220**. In other examples, however, one or more of the plurality of windows **224** can be oriented non-planar to the EO camera **216** and/or the IR cameras **220**. In this example, each of the windows **224** has an anti-reflection coating. Preferably, the windows positioned in front of the EO camera **216** are formed of a substance that is translucent to visible light, and the windows positioned in front of the IR cameras **220** are formed of a substance that is transparent to one or more IR wavelengths. For example, at least two windows of the plurality of windows **224** may be germanium windows (e.g., manufactured by Edmund Optics). In this example, the camera module **208** includes two window retention plates **228**, one window retention plate **228** for each of the pairs of openings **240**. Thus, in this example, each of the two retention plates **228** has a pair of apertures **244**, each aperture **244** sized and arranged to be aligned with a corresponding one of the openings **240** and a corresponding one of the windows **224** when the two retention plates **228** are coupled to the frame **212**. In this example, each of the two retention plates **228** is coupled to the frame **212** by disposing each of the retention plates **228** in one of the mounting cavities **246** formed in the frame **212** and inserting a plurality of fasteners (only one of which is shown in FIG. **2G**) through the frame **212** and the respective retention plate **228**.

[0059] In this example, and as best illustrated in FIGS. **2I-2L**, the frame **212** has an outer surface that is curved, creating the appearance that the windows **224** are curved as well (even though they are flat). In this example, each window retention plate **228** has an outer surface that is also curved and is flush with the outer surface of the frame **212** (as well as a surface of the hull cap **108**). Further, it will be appreciated that the camera module **208** may also include a plurality of first retaining rings **248** and a plurality of second retaining rings **249**. The first retaining rings **248**, which in this example take the form of shock-resistant threaded retention rings (e.g., manufactured by Thorlabs, inc.), are seated in a channel formed in the frame **212** at a position surrounding the openings **240**, respectively. In turn, the first retaining rings **248** are disposed so as to sealingly engage and retain the rear surface of the windows **224**, respectively. Meanwhile, the second retaining rings **249** in this example also take the form of shock-resistant threaded retention rings but are seated in a channel formed in one of the window

retention plates **228** at a position surrounding a corresponding one of the apertures **244**. In turn, the second retaining rings **249** are disposed so as to sealingly engage and retain the front surface of the windows **224**, respectively. Beneficially, the first and second retaining rings **248**, **249** are configured to use in connection with windows **224** of different thicknesses, such that windows **224** of different thicknesses can be employed in the vision system **200** as needed.

[0060] The camera module **208** also generally includes an EO bracket **250** and an IR bracket **254**. As best illustrated in FIGS. **2H** and **2N**, the EO bracket **250** in this example has a base **256**, a camera support **258**, a first wall **260**, and a second wall **262**. The camera support **258** is coupled to and extends outward (upward in FIGS. **2H** and **2N**) from a central portion of the base **256**. As such, the camera support **258** is positioned to receive and support the EO camera **216**. The first wall **260** is coupled to and extends outward (upward in FIGS. **2H** and **2N**) from a first end of the base **256**, whereas the second wall **262** is coupled to and extends outward (upward in FIGS. **2H** and **2N**) from a second end of the base **256**. In turn, the second wall **262** is disposed opposite and faces the first wall **260**, and each of the first and second walls **260**, **262** is oriented along a respective longitudinal axis **264** perpendicular to a transverse axis **266** along which the base **256** is oriented.

[0061] Meanwhile, as best illustrated in FIGS. **2H** and **20**, the IR bracket **254** in this example has a base **270**, a first camera support **272**, and a second camera support **274**. The first camera support **272** is generally configured to receive and support one of the IR cameras **220**, whereas the second camera support **272** is generally configured to receive and support the other IR camera **220**. More particularly, the first camera support **272** is coupled to a first end of the base **270**, whereas the second camera support **272** is coupled to a second end of the base **270** opposite the first camera support **272**. Each of the first and second camera supports **272**, **274** is defined by a pair of walls, a first wall **276** that is oriented along a respective longitudinal axis **278** and a second wall **280** that is, like the base **270**, oriented along a transverse axis **282** that is perpendicular to the longitudinal axes **278**. As best illustrated in FIG. **20**, the IR bracket **254** in this example also includes a pair of openings **284**, one in each of the second walls **280**, sized to receive and retain a respective one of the IR cameras **220** therein.

[0062] Turning now to FIGS. **2F**, **2H**, and **2T**, the EO camera **216** is coupled to the EO bracket **250** via fasteners **286** disposed in the camera support **258** and the rear side of the EO camera **216**. In turn, the EO camera **216** is centrally located in the EO bracket **250**, with a first cavity **288** defined between the EO camera **216** and the first wall **260** and a second cavity **290** defined between the EO camera **216** and the second wall **262**. The EO bracket **250** is also coupled to the frame **212** so as to retain the EO camera **216** in position immediately adjacent two of the openings **240**. In some examples, the EO camera **216** will be fixed in this position, while in other examples, the position of the EO camera **216** can be adjusted as needed. More particularly, in this example, the EO bracket **250** is coupled to the face plate **232** via a plurality of fasteners **292** carried by an outer portion of the EO bracket **250** and inserted into apertures formed in the rear surface of the face plate **232**. In other examples, however, the EO bracket **250** can be coupled to the face plate **232** and/or the cap **236** in a different manner.

[0063] With reference still to FIGS. **2F**, **2H**, and **2T**, the IR cameras **220** are coupled to the IR bracket **254** such that the IR cameras **220** are carried by and extend outward (downward in FIGS. **2F**, **2H**, and **2T**) from the base **270**. In this example, the IR cameras **220** are coupled to the IR bracket **254** by way of the openings **284**, through which the IR cameras respectively extend, and a plurality of fasteners (not shown) inserted into a retaining ring (not shown) and apertures formed in the second walls **280** (see FIG. **20**). In turn, the IR cameras **220** are located at the first and second ends of the base **270**.

[0064] As also illustrated in FIGS. **2F**, **2H**, and **2T**, the IR bracket **254** is coupled to the EO bracket **250** such that the IR cameras **220** are coupled to the EO camera **216** in a single unit. In this example, the IR bracket **254** is coupled to the EO bracket **250** by disposing the first and second camera supports **272**, **274** in the first and second cavities **288**, **290**, respectively, and inserting a plurality of fasteners **294** (see FIG. **2H**) into a plurality of apertures respectively formed in each of the first walls **276**. In turn, the IR bracket **254** is partially disposed in the EO bracket **250**, and the IR cameras **220** are disposed in the first and second cavities **288**, **290**, with the IR cameras **220** substantially horizontally aligned with the EO camera **216** and one IR camera **220** on each side of the EO camera **216**, as best illustrated in FIG. **2T**. Moreover, the axes **264**, **278** will be parallel (or substantially parallel) and the axes **266**, **282** will be parallel (or substantially parallel). Further, by virtue of being coupled to the EO bracket **250**, the IR bracket **254** is also coupled to the frame **212** and is configured to retain the IR cameras **220** in position immediately adjacent the other two openings **240**. In some examples, the IR cameras **220** will be fixed in this position, while in other examples, the position of the IR cameras **220** can be adjusted as needed. It will be appreciated that as referenced herein, objects (e.g., cameras, sensors) may be immediately adjacent to other objects (e.g., openings) without directly contacting them. For example, a sensor may be immediately adjacent to an opening when the edges of the opening do not obscure the FOV of the sensor.

[0065] As discussed above, the vision system **200** also includes the electronics plate **209**, which is coupled to both the housing **204** and the camera module **208**. In this example, the electronics plate **209** is removably coupled to the housing **204** via a plurality of latches **302**. In other examples, however, the electronics plate **209** can be removably coupled to the housing **204** in a different manner or can be permanently coupled (e.g., welded) to the housing **204**. In this example, the electronics plate **209** includes various electrical components for the camera module **208**, including, for example, a heat sink **304** for dissipating heat generated by the electrical components of the vision system **200** (e.g., the camera **216** and/or the cameras **220**), one or more fans **308** arranged to direct air into the heat sink **304**, an autonomous computer **312**, and a communication module **316**. The electronics plate **209** can include other electrical or mechanical components as well. In other examples, however, one or more of these electrical components (e.g., the computer **312**, the communication module **316**) can be located remotely from the vision system **200**.

[0066] When the camera module **208** is coupled to the housing **204**, it will be appreciated that the baseline distance between pairs of the dual EO image sensors and the EO/IR cameras **216**, **220** is well-toleranced and is maximized as much as possible. In turn, the delta is consistent and the

vision system **200** enables downstream ranging. When the maritime vehicle **100** is in use, and the vision system **200** is operational, the vision system **200** can, for example, have the FOV illustrated in FIG. 2V. Other FOVs are described in greater detail in described in U.S. Provisional Application No. 63/742,533, titled “Perception Hardware Configurations for Maritime Vehicle” and filed Jan. 7, 2025, the contents of which are hereby incorporated by reference herein.

[0067] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

1. A camera module for a maritime vehicle, the camera module comprising:

- a frame comprising first and second openings;
- first and second windows disposed in the frame immediately adjacent the first and second openings, respectively;
- a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings;
- a stereo camera disposed within the frame, the stereo camera disposed immediately adjacent the first opening; and
- an infrared (IR) camera disposed within the frame, the IR camera disposed immediately adjacent the second opening.

2. The camera module of claim **1**, wherein the first and second windows are germanium windows.

3. The camera module of claim **1**, wherein the frame has an outer surface that is curved, wherein the retention plate has an outer surface that is curved and is flush with the outer surface of the frame, and wherein the first and second windows are flat.

4. (canceled)

5. The camera module of claim **1**, wherein the frame is defined by a face plate and a cap coupled to the face plate, wherein each of the stereo camera and the IR camera is at least partially disposed between the face plate and the cap.

6. The camera module of claim **1**, further comprising a stereo bracket coupled to the frame, wherein the stereo camera is carried by the stereo bracket.

7. The camera module of claim **6**, further comprising an IR bracket coupled to the frame, wherein the IR camera is carried by the IR bracket.

8. The camera module of claim **7**, wherein the IR bracket is coupled to the stereo bracket.

9. The camera module of claim **8**, wherein the IR bracket is partially disposed in the stereo bracket such that the IR camera is horizontally aligned with the stereo camera.

10. The camera module of claim **1**, wherein the retention plate is disposed in a mounting cavity formed in a front side of the frame.

11. The camera module of claim **1**, further comprising a retaining ring configured to secure the IR camera to the frame.

12. The camera module of claim **1**, wherein the stereo camera includes a first electro-optical (EO) image sensor disposed immediately adjacent the first opening and a second EO image sensor disposed immediately adjacent a third opening formed in the frame.

13. The camera module of claim **1**, wherein the IR camera is a passive camera excluding any active emission system.

14. A camera module for a maritime vehicle, the camera module comprising:

- a frame comprising first and second openings;
- first and second windows disposed in the frame immediately adjacent the first and second openings, respectively;
- a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings; and
- a stereo camera disposed within the frame, the stereo camera including a first electro-optical (EO) image sensor disposed immediately adjacent the first opening and a second EO image sensor disposed immediately adjacent the second opening.

15. (canceled)

16. The camera module of claim **14**, wherein the frame has an outer surface that is curved, wherein the retention plate has an outer surface that is curved and is flush with the outer surface of the frame, and wherein the first and second windows are flat.

17. (canceled)

18. The camera module of claim **14**, wherein the frame is defined by a face plate and a cap coupled to the face plate, wherein the stereo camera is at least partially disposed between the face plate and the cap.

19. The camera module of claim **14**, further comprising a stereo bracket coupled to the frame, wherein the stereo camera is carried by the stereo bracket.

20. The camera module of claim **14**, further comprising: an infrared (IR) camera disposed within the frame, the IR camera disposed immediately adjacent a third opening formed in the frame; and

an IR bracket coupled to the frame, wherein the IR camera is carried by the IR bracket.

21. (canceled)

22. (canceled)

23. (canceled)

24. A vision system for a maritime vehicle, comprising: a housing adapted to be mounted to a hull of the maritime vehicle; and

- a camera module carried by the housing;
- a frame comprising first and second openings;
- first and second windows disposed in the frame immediately adjacent the first and second openings, respectively;
- a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings;
- a stereo camera disposed within the frame, the stereo camera disposed immediately adjacent the first opening; and
- an infrared (IR) camera disposed within the frame, the IR camera disposed immediately adjacent the second opening.

25. The vision system of claim **24**, wherein the housing is a fiberglass housing.

26. (canceled)

27. The vision system of claim 24, wherein the frame has an outer surface that is curved, [and] wherein the retention plate has an outer surface that is curved and is flush with the outer surface of the frame, and wherein the first and second windows are flat.

28. (canceled)

29. The vision system of claim 24, wherein the frame is defined by a face plate and a cap coupled to the face plate, wherein each of the stereo camera and the IR camera is at least partially disposed between the face plate and the cap.

30. The vision system of claim 24, further comprising: a stereo bracket coupled to the frame, wherein the stereo camera is carried by the stereo bracket; and an IR bracket coupled to the frame and to the stereo bracket, wherein the IR camera is carried by the IR bracket.

31. (canceled)

32. (canceled)

33. (canceled)

34. The vision system of claim 24, further comprising: an electronics plate coupled to the camera module; and a heat sink coupled to the electronics plate at a position adjacent the first and second openings.

35. (canceled)

36. (canceled)

37. (canceled)

38. (canceled)

39. A maritime vehicle, comprising:

a hull;

a hull cap coupled to the hull;

a housing sealingly mounted to a front portion of the hull cap; and

a camera module carried by the housing;

a frame comprising first and second openings;

first and second windows disposed in the frame immediately adjacent the first and second openings, respectively;

a retention plate coupled to the frame and configured to retain the first and second windows in the frame immediately adjacent the first and second openings; a stereo camera disposed within the frame, the stereo camera disposed immediately adjacent the first opening; and

an infrared (IR) camera disposed within the frame, the IR camera disposed immediately adjacent the second opening.

40. The maritime vehicle of claim 39, wherein the housing is a fiberglass housing.

41. (canceled)

42. The maritime vehicle of claim 39, wherein the frame has an outer surface that is curved, wherein the retention plate has an outer surface that is curved and is flush with the outer surface of the frame, and wherein the first and second windows are flat.

43. (canceled)

44. (canceled)

45. The maritime vehicle of claim 39, further comprising: a stereo bracket coupled to the frame, wherein the stereo camera is carried by the stereo bracket; and an IR bracket coupled to the frame and to the stereo bracket, wherein the IR camera is carried by the IR bracket.

46. (canceled)

47. (canceled)

48. (canceled)

49. (canceled)

50. (canceled)

51. (canceled)

52. (canceled)

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